

1

NUMBER AND ALGEBRA

PRODUCTS AND FACTORS

Algebra uses symbols and formulas to describe number patterns and relationships in our natural and physical world. Algebra has helped people to unlock some of the secrets of the universe, such as showing that mass and energy are related to one another by the equation $E = mc^2$. Algebra is used to better understand climate change, population growth and economic growth. Algebraic modelling is used to predict the behaviour of bushfires, which may help save lives and reduce property damage.



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Chapter outline

	Working mathematically				
1.01 Adding and subtracting terms	U	F		R	
1.02 Multiplying and dividing terms	U	F		R	
1.03 Adding and subtracting algebraic fractions	U	F			
1.04 Multiplying and dividing algebraic fractions	U	F			
1.05 Expanding expressions	U	F			
1.06 Factorising expressions	U	F			
1.07 Expanding binomial products	U	F		R	C
1.08 Perfect squares*	U	F		R	C
1.09 Difference of 2 squares*	U	F		R	C
1.10 Mixed expansions*	U	F		R	C
1.11 Factorising special binomial products*	U	F			
1.12 Factorising quadratic expressions $x^2 + bx + c$*	U	F		R	
1.13 Factorising quadratic expressions $ax^2 + bx + c$*	U	F		R	
1.14 Mixed factorisations*	U	F		R	
1.15 Factorising algebraic fractions*	U	F		R	

*** PATH**

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

binomial An algebraic expression that consists of 2 terms, for example, $4a + 9$, $3 - y$, $x^2 - 4x$

binomial product An algebraic expression showing 2 or more binomials multiplied together, for example, $(x + 9)(3x - 4)$.

expand To rewrite an expression such as $5(2k - 6)$ without grouping symbols; for example, $5(2k - 6)$ expands to $10k - 30$

factorise To rewrite an expression with grouping symbols, by taking out the highest common factor; factorising is the opposite of expanding; for example, $9r^2 + 36r$ factorised is $9r(r + 4)$

highest common factor (HCF) The largest term that is a factor of 2 or more terms, for example, the HCF of $9r^2 + 36r$ is $9r$

perfect square A square number or an algebraic expression that represents one, for example, 64, $(x + 9)^2$

quadratic expression An algebraic expression in which the highest power of the variable is 2, for example, $2x^2 + 5x - 3$ or $x^2 + 2$

quadratic trinomial A quadratic expression that consists of 3 terms, for example, $x^2 + 2x + 6$

Quiz
Wordbank 1

Videos (18):

1.01 Adding and subtracting terms
1.02 Multiplying terms • Dividing terms
1.03 Adding and subtracting algebraic fractions
1.04 Algebraic fractions
1.05 Expanding expressions 1 • Expanding expressions 2
1.06 HCF of algebraic terms • Factorising expressions
1.07 Expanding binomial products 1 • Expanding binomial products 2
1.08 Special binomial products
1.09 Special binomial products • Difference of 2 squares
1.11 Factorisation by grouping in pairs • Difference of 2 squares
1.12 Factorising quadratic expressions 1
1.13 Factorising quadratic expressions (Advanced) • Factorising quadratic expressions 2
1.14 Mixed factorisations

PhET interactives (2):

1.01 Expression exchange
1.05 Area model Algebra

Quizzes (6):

- Wordbank 1
- SkillCheck 1
- Mental skills 1A
- Mental skills 1B
- Language of maths 1
- Test yourself 1

Skillsheets (4):

SkillCheck Algebraic expressions
1.01, 1.05 Algebra using diagrams
1.06 HCF by factor trees • Factorising using diagrams

Worksheets (12):

1.02 Perimeter and area
1.04 Algebraic fractions
1.05 Algebra review
1.07 Area diagrams • Binomial products
1.09 Special products
1.10 Mixed expansions • Algebra 6
1.14 Mixed factorisations
1.15 Simplifying algebraic fractions • Algebra 2
Mind map Products and factors

Puzzles (17):

SkillCheck Generalised arithmetic • Substitution puzzle
1.03 Algebraic fractions
1.04 Algebraic fractions puzzle • Upside-down fractions
1.06 Factorising puzzle
1.07 Expanding brackets • Trinominoes
1.08 Factoronimoes • Perfect squares
1.09 Enough time
1.10 Expanding binomials
1.11 Grouping • Difference of 2 perfect squares
1.12 Perfect squares
1.13 Trinominoes
1.14 Factorising puzzle (Advanced) • Factoronimoes • Products and factors squaresaw
Language of maths Algebra crossword

Technology (2):

1.07 Expanding binomials
1.12 Factorising trinomials

Spreadsheets (2):

1.07 Expanding binomials
1.12 Factorising trinomials

Presentation (1):

1.03 Adding and subtracting fractions

Nelson MindTap

To access resources above, visit
cengage.com.au/nelsonmindtap

In this chapter you will:

- add, subtract, multiply and divide algebraic expressions
- add, subtract, multiply and divide simple algebraic fractions
- expand and factorise algebraic expressions, including binomial products
- (PATH) recognise and expand special binomial products that are perfect squares or difference of 2 squares
- (PATH) factorise simple (monic) quadratic expressions
- (PATH) factorise an expression with 4 terms by grouping in pairs
- (PATH) factorise non-monic quadratic expressions, including those that are perfect squares or difference of 2 squares
- (PATH) factorise and simplify algebraic fractions

SkillCheck ANSWERS ON P. 608

1 Write an **algebraic expression** for each statement.

a The cost of p books at \$$Q$ each

b The selling price after a GST of \$$t$ is added to the marked price of \$40

c The number of metres in h km

d The average of m, p, v and w

e The number of dollars in y cents

2 If $f = 5$, $g = -4$ and $r = 2$, then evaluate each expression.

a $2f + g$ **b** $8 + rf$ **c** $f^2 - r^2$

d $3g^2 - 5$ **e** $r(f - g)$ **f** $\frac{5r + f}{3}$

3 If the perimeter of a rectangle with length l and width w has the formula $P = 2l + 2w$, find the perimeter of a rectangle with length 4.7 cm and width 2.5 cm.

4 Simplify each expression.

a $5n + 6n$ **b** $9h - 3h$ **c** $10pq - qp$ **d** $x^2 + 4x^2$

e $4 \times w$ **f** $8 \times 5x$ **g** $2g \times 3h$ **h** $3y \times 6y$

i $10x \div 2$ **j** $6b \div b$ **k** $\frac{16xy}{8x}$ **l** $\frac{15n^2}{25n}$

5 Evaluate each expression.

a $\frac{2}{5} + \frac{1}{3}$ **b** $\frac{3}{4} - \frac{2}{5}$ **c** $\frac{5}{8} \times \frac{1}{2}$ **d** $\frac{4}{5} \div \frac{2}{7}$

e $\frac{6}{11} \times \frac{5}{12}$ **f** $\frac{7}{8} + \frac{1}{3}$ **g** $\frac{4}{9} \div \frac{3}{10}$ **h** $\frac{5}{6} - \frac{3}{4}$

i $1\frac{3}{4} + 2\frac{5}{7}$ **j** $3\frac{1}{2} \times 2\frac{3}{5}$ **k** $2\frac{4}{9} \div 1\frac{5}{6}$ **l** $5\frac{1}{4} - 2\frac{2}{3}$

Quiz
SkillCheck 1

Skillsheet
Algebraic expressions

Puzzles
Generalised arithmetic

Substitution puzzle

1.01 Adding and subtracting terms

Video
Adding and subtracting terms

Skillsheet
Algebra using diagrams

Interactive
Expression exchange

Adding and subtracting terms

Only **like terms** can be added or subtracted

Example 1

Simplify each expression.

a $5r + 4s - 7r - 2s$

b $11u^2 - 5u + 3u - 7u^2$

Solution

a
$$\begin{aligned} 5r + 4s - 7r - 2s &= 5r - 7r + 4s - 2s \\ &= -2r + 2s \\ &= 2s - 2r \end{aligned}$$

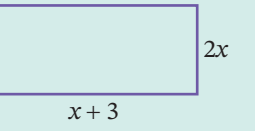
Group the like terms
$-2r$ and $2s$ cannot be simplified further
Rewrite with positive term first

b
$$\begin{aligned} 11u^2 - 5u + 3u - 7u^2 &= 11u^2 - 7u^2 - 5u + 3u \\ &= 4u^2 - 2u \end{aligned}$$

Each + and – sign belongs to the term that follows it.

Example 2

Write a simplified algebraic expression for the perimeter of this rectangle.



Solution

$$\begin{aligned} \text{Perimeter} &= x + 3 + 2x + x + 3 + 2x \\ &= x + 2x + x + 2x + 3 + 3 \\ &= 6x + 6 \end{aligned}$$

EXERCISE 1.01 ANSWERS ON P. 608

Adding and subtracting terms U F R

EXAMPLE 1

1 Simplify each algebraic expression.

a	$2fg + 3fg - 4fg$	**b**	$6uvw - uvw + 4uvw$	**c**	$3t + 3 + 5t$
d	$3x + 2y + 8y$	**e**	$6g + h - 4g$	**f**	$2fg + 3f - 4fg$
g	$7p^2 - 3 - 2p^2$	**h**	$4p - q - 3q$	**i**	$-8r + 6r + 3$
j	$-3a + 4b - 7a$	**k**	$4v + 3 + 2v + 7$	**l**	$2y + 6x + y - 3x$
m	$7p - 8q - p - q$	**n**	$8 - 3w + 7 - 2w$	**o**	$2n - 3 - 12n + 5$
p	$8e - 3f - 2e - 4f$	**q**	$2l^2 + 2l + 3l^2 - 6l$	**r**	$b - 12b^2 + 6b^2 - 15b$
s	$6p^3 - 5p^2 - 4p^3 + p^2$	**t**	$3 - 8k - 5 - 9k$	**u**	$-3 - 5d + 9 - 8d$

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2 Simplify $15a^2 + 12a + 48 - 18a$. Select the correct answer **A**, **B**, **C** or **D**.

A $24a + 48$ **B** $27a^2 + 30a$

C $15a^2 - 6a + 48$ **D** $15a^2 + 30a - 48$

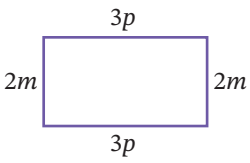
3 Use the substitutions $a = 2$, $b = 3$ and $c = 4$ to test whether each equation is correct or incorrect. R

a $20c - 12c = 8c$ **b** $9b + b = 9b^2$ **c** $7a + 3b = 10 + a + b$

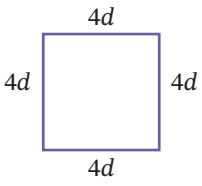
d $3a^2 + 5a^2 = 8a^2$ **e** $4ab + 3ab = 7ab$ **f** $12c - c = 12$

4 Write a simplified algebraic expression for the perimeter of each shape.

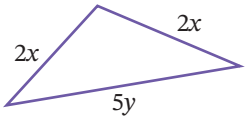
a



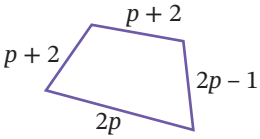
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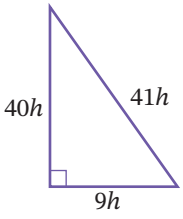
c



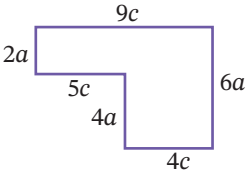
d



e



f



5 Draw a square and write an algebraic expression for its side length so that it has a perimeter of $20x$. R

6 Draw an equilateral triangle and write an algebraic expression for its side lengths so that it has a perimeter of $9d + 6$. R

7 Draw a rectangle and find algebraic expressions for its length and width so that it has a perimeter of $18a - 30$. R

8 Draw a triangle and write algebraic expressions for its side lengths so that it has a perimeter of $18a - 30$. R

EXAMPLE 2

1.02

Multiplying and dividing terms 1.02

Multiplying and dividing terms

When **multiplying and dividing terms**, multiply or divide the numbers first, then the variables.

Video
Multiplying terms

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Worksheet
Perimeter and area

Example 3

Simplify each expression.

a $3d \times 4d$ **b** $2g \times 3h \times (-5f)$

Solution

a $3d \times 4d = 3 \times 4 \times d \times d$
$= 12d^2$

b $2g \times 3h \times (-5f) = 2 \times 3 \times (-5) \times g \times h \times f$
$= -30fgh$

Video
Dividing terms

Example 4

Simplify each expression.

a $\frac{k^2m}{4km}$ **b** $35p^2 \div 15pw$

Solution

a $\frac{k^2m}{4km} = \frac{kkm}{4km}$
$= \frac{1}{4} \frac{k \cancel{km}^{1}}{\cancel{km}^{1}}$
$= \frac{k}{4}$

b $35p^2 \div 15pw = \frac{35p^2}{15pw}$
$= \frac{\cancel{35}^{7}}{\cancel{15}^{3}} \frac{\cancel{p}^{1}p}{\cancel{p}^{1}w}$
$= \frac{7p}{3w}$

EXERCISE 1.02 ANSWERS ON P. 608

Multiplying and dividing terms U F R

EXAMPLE 3

1 Simplify each expression.

a $4d \times 3e$ **b** $6v \times 4w$ **c** $11q \times 8q$

d $-5a \times (-4a)$ **e** $2y \times 3 \times 4b$ **f** $-6w \times 3w \times 2$

g $-9c \times (-4h) \times 2k$ **h** $\frac{1}{3}a \times 24bc$ **i** $-5d \times (-3e) \times (-2w)$

j $(5n)^2$ **k** $(-3r)^2$ **l** $(at)^2$

m $-15d \times \frac{2}{3}ad$ **n** $-\frac{1}{4}ek \times (-4k)$ **o** $(-2xy)^2$

EXAMPLE 4

2 Simplify each expression.

a $\frac{12x}{3}$ **b** $\frac{15y}{y}$ **c** $\frac{36a}{4a}$

d $18mn \div 6m$ **e** $8vw \div 48vw$ **f** $\frac{64abc}{16b}$

g $\frac{18g}{-3g}$ **h** $-25yz \div 5y$ **i** $\frac{9p}{-45pq}$

j $\frac{-75xyz}{-25x}$ **k** $10r^2 \div 40r$ **l** $-12hk^2 \div 2hk$

m $24ak^2 \div 8a^2$ **n** $16mwx \div 12mx$ **o** $7gh \div 21g$

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3 Simplify $\frac{9yx^2}{24xy^2}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{3x^2}{8y^2}$ **B** $\frac{3x^3}{8y^3}$ **C** $\frac{3x}{8y}$ **D** $\frac{3y}{8x}$

4 Simplify each expression.

a $2a \times 3b \times 5c$ **b** $4p \times 3 \times 2q$ **c** $40n^2 \div 5n$

d $8b^2 \div 40b^2$ **e** $6h \times 2k \times (-3)$ **f** $-2 \times 5k \times (-4k)$

g $\frac{6d}{12}$ **h** $\frac{7r}{49r}$ **i** $(-33m) \div 3$

j $\frac{-9w}{-45w}$ **k** $8ab \div 2bc$ **l** $2ef \times (-3fe)$

m $18ac^2d \div 27acd$ **n** $-3pq \times (-2q) \times (-4)$ **o** $\frac{24v^2w}{18v}$

5 Substitute $p = 5$, $q = 4$ and $r = 3$ to test whether each equation is correct or incorrect.

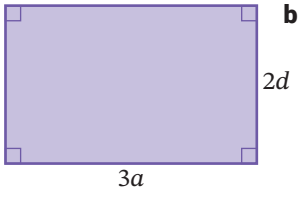
a $3 \times 4p = 12p$ **b** $q + q = q^2$ **c** $2r \times 3r = 6r^2$

d $3q^2 \times 5q^2 = 15q^2$ **e** $\frac{12p}{3p} = 4$ **f** $\frac{7qr}{qr} = 7$

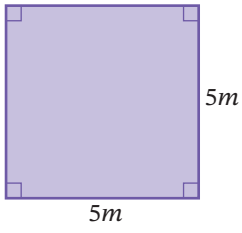
g $3r \times 5q = 15qr$ **h** $p \times q \times q = 2pq$ **i** $\frac{14r^2}{7r} = 2r$

6 Write a simplified algebraic expression for the area of each shape.

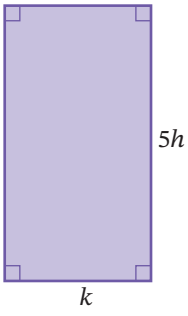
a



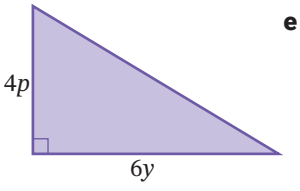
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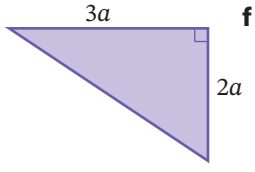
c



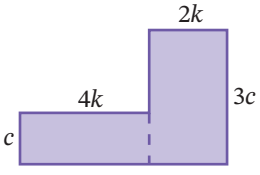
d



e

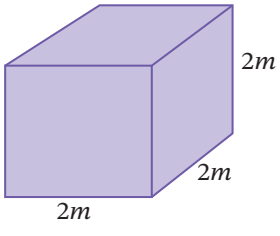


f

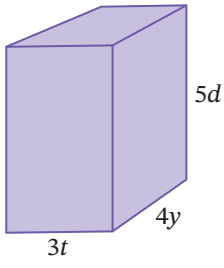


7 Find an algebraic expression for the volume of each rectangular prism.

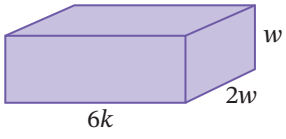
a



b



c



8 Draw a square and write an algebraic expression for its side length if it has an area of $100x^2$. R

9 Draw a rectangle and write algebraic expressions for its length and width if it has an area of $36pq$. R

Foundation Standard Complex

1.03 Adding and subtracting algebraic fractions

Puzzle
Algebraic fractions

Presentation
Adding and subtracting fractions

Video
Adding and subtracting algebraic fractions

Adding and subtracting fractions

To **add or subtract fractions**, convert them (if needed) so that they have the same denominator, then simply add or subtract the numerators.

Example 5

Simplify each expression.

a $\frac{7r}{10} - \frac{3r}{10}$ **b** $\frac{k}{3} + \frac{k}{2}$ **c** $\frac{3h}{4} + \frac{2h}{5}$

d $\frac{5m}{6} - \frac{2m}{3}$ **e** $\frac{3}{4} + \frac{7a}{10}$ **f** $\frac{5x}{8} - \frac{2w}{3}$

Solution

a $\frac{7r}{10} - \frac{3r}{10} = \frac{4r}{10}$
$= \frac{2r}{5}$

b $\frac{k}{3} + \frac{k}{2} = \frac{2 \times k}{2 \times 3} + \frac{3 \times k}{3 \times 2}$
$= \frac{2k}{6} + \frac{3k}{6}$
$= \frac{5k}{6}$

c $\frac{3h}{4} + \frac{2h}{5} = \frac{5 \times 3h}{5 \times 4} + \frac{4 \times 2h}{4 \times 5}$
$= \frac{15h}{20} + \frac{8h}{20}$
$= \frac{23h}{20}$

d $\frac{5m}{6} - \frac{2m}{3} = \frac{15m}{18} - \frac{12m}{18}$
$= \frac{3m}{18}$
$= \frac{m}{6}$

e $\frac{3}{4} + \frac{7a}{10} = \frac{15}{20} + \frac{14a}{20}$
$= \frac{15 + 14a}{20}$

or $\frac{5m}{6} - \frac{2m}{3} = \frac{5m}{6} - \frac{4m}{6}$
$= \frac{m}{6}$

f $\frac{5x}{8} - \frac{2w}{3} = \frac{15x}{24} - \frac{16w}{24}$
$= \frac{15x - 16w}{24}$

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EXERCISE 1.03 ANSWERS ON P. 608

Adding and subtracting algebraic fractions U F

EXAMPLE 5

1.03

1 Simplify each expression.

a $\frac{w}{4}+\frac{2w}{4}$ **b** $\frac{4k}{8}+\frac{7k}{8}$ **c** $\frac{7m}{10}-\frac{2m}{10}$ **d** $\frac{x}{3}+\frac{2x}{3}$

e $\frac{4}{q}+\frac{5}{q}$ **f** $\frac{5}{3d}-\frac{2}{3d}$ **g** $\frac{4r}{w}+\frac{r}{w}$ **h** $\frac{5}{c}-\frac{2}{c}$

i $\frac{4t}{3}+\frac{s}{3}$ **j** $\frac{11y}{2h}-\frac{7y}{2h}$ **k** $\frac{6}{a}-\frac{5}{a}$ **l** $\frac{8}{5e}+\frac{7}{5e}$

m $\frac{p}{z}+\frac{3}{z}$ **n** $\frac{5u}{8g}-\frac{3u}{8g}$ **o** $\frac{4}{9f}-\frac{1}{9f}$ **p** $\frac{7e}{10}-\frac{3e}{10}$

q $\frac{5a}{6}+\frac{5a}{6}$ **r** $\frac{5}{2d}+\frac{3}{2d}$ **s** $\frac{7}{5k}+\frac{13}{5k}$ **t** $\frac{12}{4a}-\frac{8}{4a}$

2 Simplify the expression $\frac{9}{4k}-\frac{3}{4k}$. Select **A**, **B**, **C** or **D**.

A $\frac{6}{4k}$ **B** $\frac{3}{4k}$ **C** $\frac{3}{2k}$ **D** $\frac{6}{k}$

3 Simplify each expression.

a $\frac{x}{3}+\frac{x}{4}$ **b** $\frac{s}{3}-\frac{s}{7}$ **c** $\frac{h}{5}+\frac{h}{3}$ **d** $\frac{m}{7}-\frac{m}{2}$

e $\frac{w}{4}+\frac{w}{5}$ **f** $\frac{5t}{4}-\frac{2t}{5}$ **g** $\frac{2p}{5}+\frac{p}{3}$ **h** $\frac{5r}{2}-\frac{5r}{3}$

i $\frac{5y}{6}+\frac{2y}{3}$ **j** $\frac{3m}{4}-\frac{5m}{12}$ **k** $\frac{5a}{9}-\frac{a}{6}$ **l** $\frac{3y}{10}+\frac{7y}{15}$

m $\frac{3c}{2}-\frac{c}{5}$ **n** $\frac{2d}{11}-\frac{r}{3}$ **o** $\frac{3h}{5}+\frac{2d}{3}$ **p** $\frac{5}{6}+\frac{4w}{5}$

q $\frac{2a}{7}+\frac{3}{4}$ **r** $\frac{4e}{7}-\frac{2e}{5}$ **s** $\frac{3m}{2}-\frac{5n}{11}$ **t** $\frac{2k}{5}+\frac{m}{6}$

4 Simplify the expression $\frac{5m}{12}-\frac{3m}{8}$. Select **A**, **B**, **C** or **D**.

A $\frac{2m}{4}$ **B** $\frac{m}{24}$ **C** $\frac{m}{2}$ **D** $\frac{3m}{20}$

5 Simplify the expression $\frac{4h}{9}+\frac{2d}{5}$. Select **A**, **B**, **C** or **D**.

A $\frac{36h+10d}{45}$ **B** $\frac{4h+2d}{14}$ **C** $\frac{20h+18d}{45}$ **D** $\frac{9h+18d}{14}$

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1.04 Multiplying and dividing algebraic fractions

Worksheet
Algebraic fractions

Puzzles
Algebraic fractions puzzle

Upside-down fractions

Video
Algebraic fractions

Multiplying and dividing algebraic fractions

- To **multiply fractions**, cancel any common factors, then multiply the numerators and denominators separately
- To **divide by a fraction** $\frac{a}{b}$, multiply by its **reciprocal** $\frac{b}{a}$

Example 6

Simplify each **product**.

a $\frac{3}{d} \times \frac{4}{c}$ **b** $\frac{4}{k} \times \frac{3k}{16}$ **c** $\frac{3x}{4} \times \frac{2x}{9}$

Solution

a $\frac{3}{d} \times \frac{4}{c} = \frac{3 \times 4}{d \times c}$

$= \frac{12}{dc}$

b $\frac{4}{k} \times \frac{3k}{16} = \frac{\cancel{4}^{1}}{\cancel{k}_{1}} \times \frac{\cancel{3k}}{\cancel{16}_{4}}$

$= \frac{3}{4}$

c $\frac{3x}{4} \times \frac{2x}{9} = \frac{\cancel{3x}^{1}}{\cancel{4}_{2}} \times \frac{\cancel{2x}^{1}}{\cancel{9}_{3}}$

$= \frac{x^2}{6}$

Video
Algebraic fractions

Example 7

Simplify each quotient.

a $\frac{3}{h} \div \frac{4}{k}$ **b** $\frac{xy}{5} \div \frac{3x}{25}$

Solution

a $\frac{3}{h} \div \frac{4}{k} = \frac{3}{h} \times \frac{k}{4}$

$= \frac{3k}{4h}$

The reciprocal of $\frac{4}{k}$ is $\frac{k}{4}$.

b $\frac{xy}{5} \div \frac{3x}{25} = \frac{\cancel{xy}^{1}}{\cancel{5}_{1}} \times \frac{\cancel{25}^{5}}{\cancel{3x}_{1}}$

$= \frac{5y}{3}$

EXERCISE 1.04 ANSWERS ON P. 608

Multiplying and dividing algebraic fractions U F

EXAMPLE 6

1 Simplify the product $\frac{3x}{5} \times \frac{w}{6}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{xw}{10}$ **B** $\frac{3xw}{11}$ **C** $\frac{3xw}{30}$ **D** $\frac{18x+5w}{30}$

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EXAMPLE 7

2 Simplify each product.

a $\frac{w}{3} \times \frac{1}{2}$ **b** $\frac{s}{5} \times \frac{t}{4}$ **c** $\frac{3}{h} \times \frac{5}{6k}$

d $\frac{4}{m} \times \frac{3}{n}$ **e** $\frac{l}{3} \times \frac{5}{f}$ **f** $\frac{1}{v} \times \frac{2}{3v}$

g $\frac{2}{x} \times \frac{3}{x}$ **h** $\frac{a}{b} \times \frac{c}{d}$ **i** $\frac{d}{e} \times \frac{e}{g}$

j $\frac{4ad}{9} \times \frac{d}{16a}$ **k** $\frac{5p}{4} \times \frac{8}{15p}$ **l** $\frac{4}{ak} \times \frac{3a}{5k}$

m $\frac{2}{3} \times \frac{9u}{10}$ **n** $\frac{u}{3} \times \frac{3}{u}$ **o** $\frac{3z}{r} \times \frac{2r}{9dz}$

3 Simplify each quotient.

a $\frac{r}{2} \div \frac{r}{5}$ **b** $\frac{m}{6} \div \frac{n}{3}$ **c** $\frac{h}{2} \div \frac{h}{8}$

d $\frac{q}{4} \div \frac{3}{4}$ **e** $\frac{3y}{5} \div \frac{2y}{d}$ **f** $\frac{4t}{9} \div \frac{3t}{5}$

g $\frac{3}{e} \div \frac{5}{e}$ **h** $\frac{3}{2a} \div \frac{b}{6a}$ **i** $\frac{5m}{n} \div \frac{2m}{3n}$

j $\frac{h}{k} \div \frac{k}{h}$ **k** $\frac{8w}{3x} \div \frac{2w}{9x}$ **l** $\frac{3s}{4} \div \frac{6s}{11}$

m $\frac{t}{3} \div \frac{3t}{5u}$ **n** $\frac{3e}{7g} \div \frac{e}{14g}$ **o** $\frac{xh}{5} \div \frac{3h}{15}$

4 Simplify the quotient $\frac{3}{4m} \div \frac{9n}{8m}$. Select **A**, **B**, **C** or **D**.

A $\frac{3n}{2}$ **B** $\frac{32m^2}{27n}$ **C** $\frac{36m}{24n}$ **D** $\frac{2}{3n}$

5 Simplify each expression.

a $\frac{3p}{2} \div \frac{p^2}{4}$ **b** $\frac{2w}{7} \div \frac{5w}{6}$ **c** $\frac{8y}{5} \times \frac{3}{32y^2}$

d $\frac{5}{2g} \div \frac{2}{g}$ **e** $\frac{xy}{z} \times \frac{5}{y}$ **f** $\frac{a}{b} \times \frac{c}{b} \times \frac{b}{3a}$

g $5 \div \frac{3}{5b}$ **h** $\frac{4t}{9} \div 3$ **i** $\frac{5mn}{2d} \times \frac{4d}{n} \times \frac{1}{15mn}$

j $5p \div \frac{7}{10pt}$ **k** $\frac{s}{3} \div \frac{5s}{2} \times \frac{3s}{7}$ **l** $\frac{7}{h} \times \frac{1}{42h}$

m $\frac{5ty}{k} \times \frac{5ky}{t}$ **n** $\frac{6}{r} \times \frac{5r}{9} \div \frac{15}{yh}$ **o** $\frac{6cf}{q} \div \frac{5}{qf} \div \frac{10c}{a}$

6 Simplify the expression $\frac{4ab}{5} \div \frac{a}{2b} \div \frac{8b}{15}$. Select **A**, **B**, **C** or **D**.

A 3 **B** $3b$ **C** $\frac{3b}{5}$ **D** $\frac{3b}{a}$

Quiz
Mental skills 1A

Mental skills 1A: Maths without calculators ANSWERS ON P. 609

Multiplying and dividing by 5, 15, 25 and 50

It is easier to multiply or divide a number by 10 than by 5. So whenever we multiply or divide a number by 5, we can double the 5 (to make 10) and then adjust the first number.

1 Study each example.

a **To multiply by 5**, halve the number, then multiply by 10.

$18 \times 5 = 18 \times \frac{1}{2} \times 10 \quad (\text{or } 9 \times 2 \times 10)$

$= 9 \times 10$

$= 90$

b **To multiply by 50**, halve the number, then multiply by 100.

$26 \times 50 = 26 \times \frac{1}{2} \times 100 \quad (\text{or } 13 \times 2 \times 100)$

$= 13 \times 100$

$= 1300$

c **To multiply by 25**, quarter the number, then multiply by 100.

$44 \times 25 = 44 \times \frac{1}{4} \times 100 \quad (\text{or } 11 \times 4 \times 25)$

$= 11 \times 100$

$= 1100$

d **To multiply by 15**, halve the number, then multiply by 30.

$8 \times 15 = 8 \times \frac{1}{2} \times 30 \quad (\text{or } 4 \times 2 \times 15)$

$= 4 \times 30$

$= 120$

e **To divide by 5**, divide by 10 and double the answer. We do this because there are two 5s in every 10.

$140 \div 5 = 140 \div 10 \times 2$

$= 14 \times 2$

$= 28$

f **To divide by 50**, divide by 100 and double the answer. This is because there are two 50s in every 100.

$400 \div 50 = 400 \div 100 \times 2$

$= 4 \times 2$

$= 8$

g **To divide by 25**, divide by 100 and multiply the answer by 4. This is because there are four 25s in every 100.

$600 \div 25 = 600 \div 100 \times 4$

$= 6 \times 4$

$= 24$

h **To divide by 15**, divide by 30 and double the answer. This is because there are two 15s in every 30.

$240 \div 15 = 240 \div 30 \times 2$

$= 8 \times 2$

$= 16$

2 Now evaluate each expression.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	28×5	**h**	12×25
i	12×15	**j**	22×35	**k**	$90 \div 5$	**l**	$170 \div 5$
m	$230 \div 5$	**n**	$1300 \div 50$	**o**	$900 \div 50$	**p**	$300 \div 25$
q	$1000 \div 25$	**r**	$360 \div 45$	**s**	$210 \div 15$	**t**	$360 \div 15$

Expanding expressions 1.05

The distributive law for expanding an expression

Multiply each term inside the brackets by the term outside

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

Skillsheet Algebra using diagrams

Worksheet Algebra review

Interactive Area model Algebra

Videos Expanding expressions 1

Expanding expressions 2

Example 8

Expand each expression.

a $3g(h - 2)$ **b** $r(10 - 4r)$

c $-(y + 4)$ **d** $-3(2m - 7)$

Solution

a $3g(h - 2) = 3g \times h - 3g \times 2$

$= 3gh - 6g$

b $r(10 - 4r) = r \times 10 - r \times 4r$

$= 10r - 4r^2$

c $-(y + 4) = -1(y + 4)$

$= -y + (-4)$

$= -y - 4$

d $-3(2m - 7) = -3 \times 2m - (-3) \times 7$

$= -6m - (-21)$

$= -6m + 21$

Example 9

Expand and simplify by collecting like terms.

a $4(3p + 2) - 5p$ **b** $5(2e - 3) - 4(1 - 5e)$ **c** $7(n - 3) + n(n - 1)$

Solution

a $4(3p + 2) - 5p = 4 \times 3p + 4 \times 2 - 5p$ Expanding
$= 12p + 8 - 5p$
$= 12p - 5p + 8$ Collecting like terms to simplify
$= 7p + 8$

b $5(2e - 3) - 4(1 - 5e) = 5 \times 2e - 5 \times 3 - 4 \times 1 - (-4) \times 5e$ Expanding
$= 10e - 15 - 4 - (-20e)$
$= 10e - 19 + 20e$
$= 30e - 19$ Collecting like terms to simplify

c $7(n - 3) + n(n - 1) = 7n - 21 + n^2 - n$
$= n^2 + 7n - n - 21$ It's conventional to place n^2 first
$= n^2 + 6n - 21$ Collecting like terms to simplify

EXERCISE 1.05 ANSWERS ON P. 609

Expanding expressions U F

EXAMPLE 8

1 Expand each expression.

a $4(h + 6)$ **b** $5(3 + t)$ **c** $3(2b + 7)$
d $5(8 + 3x)$ **e** $x(x + 9)$ **f** $p(5 + p)$
g $3r(2r + 1)$ **h** $4y(1 + 3y)$ **i** $10e(2e + 4f)$

2 Expand each expression.

a $3(t - 2)$ **b** $7(5 - d)$ **c** $8(8g - 3)$
d $w(w - 7)$ **e** $a(a - 1)$ **f** $6h(3h - 1)$
g $4x(3x - 1)$ **h** $5x(2x - 4y)$ **i** $3a(4b - 7a)$

3 Expand $-5u(8 + 2u)$. Select the correct answer **A**, **B**, **C** or **D**.

A $3u - 3u^2$ **B** $-40u + 10u^2$ **C** $-40u - 10u^2$ **D** $-40u - 3u^2$

4 Use the substitution $x = 5$ to test whether each equation is correct or incorrect.

a $3(x + 3) = 3x + 6$ **b** $x(4 - 2x) = 4x - 2x^2$

5 Expand each expression.

a $-(a - 5)$ **b** $-(a + 5)$ **c** $-2(x + 6)$
d $-2(x - 6)$ **e** $-(11 + w)$ **f** $-(11 - w)$
g $-y(y + 9)$ **h** $-x(8 - m)$ **i** $-4t(t - 8)$

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6 Expand $-k(3k - 7)$. Select **A**, **B**, **C** or **D**.

A $-2k^2 - 8k$ **B** $-3k^2 + 7k$ **C** $-3k^2 - 7k$ **D** $-3k + 7k^2$

7 Expand $3(2 - 7y) + 5y$. Select **A**, **B**, **C** or **D**.

A $6 - 26y$ **B** $6 + 16y$ **C** $6 - 5y$ **D** $6 - 16y$

EXAMPLE 9

8 Expand and simplify each expression.

a	$5(3m + 2) + 4m$	**b**	$3(1 - 5e) + 6e$	**c**	$4w - 2(5 + 2w)$
d	$8 - 5(2x - 7)$	**e**	$5(2a + 3) + 4(a + 7)$	**f**	$3(2d + 3) - 5(d + 4)$
g	$7(3g - 1) + 4(2g - 3)$	**h**	$4(3 - 4w) - 2(w - 5)$	**i**	$5(6c - 3) - 3(4 - 3c)$
j	$t(t + 4) + 3(t + 4)$	**k**	$4(3 + h) + h(7 - 2h)$	**l**	$6(2e - 1) - (5 - 3e)$
m	$3x(2x + 5) + 4(2x + 5)$	**n**	$v(2v + 3) - 6(v + 1)$	**o**	$3(1 - 2w) - w(2 - w)$
p	$2y(3y - 7) - 5(3y - 7)$	**q**	$4(3x + 2y - 1) - (x - 1)$	**r**	$4x(3x - 2y + 1) - (x + 1)$

9 Expand $3(1 - 4k) - 2(3k + 7)$. Select **A**, **B**, **C** or **D**.

A $17 - 18k$ **B** $11 - 6k$ **C** $-11 - 18k$ **D** $-17 + 6k$

Factorising expressions 1.06

The highest common factor (HCF)

The **highest common factor (HCF)** or **greatest common divisor (GCD)** of 2 or more terms is the largest term that is a factor of all terms. To find the HCF of algebraic terms:

- find the HCF of the numbers
- find the HCF of the variables
- multiply them together

Skillsheets
HCF by factor trees

Factorising using diagrams

Puzzle
Factorising puzzle

Video
HCF of algebraic terms

Example 10

Find the highest common factor (HCF) of each pair of terms.

a $20x^2$ and $15xy$ **b** 16 and $12b$

Solution

Find the HCF of the numbers and the HCF of the variables. Their product is the HCF of the expression.

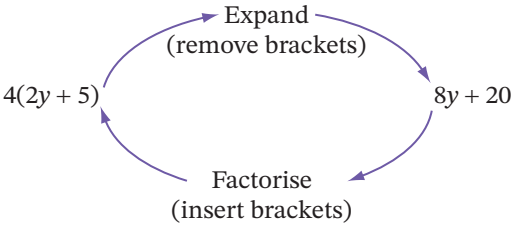
a The HCF of 20 and 15 is 5.
The HCF of x^2 and xy is x, since it is the 'largest' common part of x^2 and xy.
$\therefore$ The HCF of $20x^2$ and $15xy = 5 \times x = 5x$

b The HCF of 16 and 12 is 4.
There is **no** HCF of the variables because b is in $12b$ but not in 16.
$\therefore$ The HCF of 16 and $12b$ is 4.

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Factorising algebraic expressions

Factorising is the reverse of expanding. Factorisation breaks an expression into factors. To factorise $8y + 20$, we take out the highest common factor and **insert** brackets. The result is $4(2y + 5)$. The factors are 4 and $(2y + 5)$.



Factorising an expression

- Find the HCF of the terms and write it outside the brackets
- Divide each term by the HCF and write the answers inside the brackets

$$ab + ac = a(b + c)$$
$$ab - ac = a(b - c)$$

- To check that the factorised answer is correct, expand it

Video
Factorising expressions

Example 11

Factorise each expression.

a $8y + 16$ **b** $25b^2 - 20ab$ **c** $v(4 + w) + 2(4 + w)$

Solution

a The HCF of $8y$ and 16 is 8.

$\therefore 8y + 16 = 8 \times y + 8 \times 2$ Rewrite the expression using the HCF 8

$= 8(y + 2)$ Write the HCF at the front of the brackets

b The HCF of $25b^2$ and $20ab$ is $5b$.

$\therefore 25b^2 - 20ab = 5b \times 5b - 5b \times 4a$ Rewrite the expression using the HCF $5b$

$= 5b(5b - 4a)$ Write the HCF at the front of the brackets

c The HCF of $v(4 + w)$ and $2(4 + w)$ is $(4 + w)$.

$\therefore v(4 + w) + 2(4 + w) = (4 + w) \times v + (4 + w) \times 2$

$= (4 + w)(v + 2)$

Factorising with negative terms

Example 12

Factorise each expression.

a $-x^2 + 3x$ **b** $-a - ab$

Solution

When factorising expressions that begin with a **negative** term, we use the 'negative' HCF.

a The highest 'negative' common factor of $-x^2$ and $3x$ is $-x$.

$\therefore -x^2 + 3x = (-x) \times x + (-x) \times (-3)$ $(-x) \times (-3) = +3x$

$= (-x)[x + (-3)]$

$= -x(x - 3)$ Check your answer by expanding.

b The highest 'negative' common factor of $-a$ and ab is $-a$.

$\therefore -a - ab = -a \times 1 + (-a) \times b$ $(-a) \times b = -ab$

$= -a(1 + b)$

Factorising with negative terms

- Find the 'negative' HCF of the terms and write it outside the brackets
- Divide each term by the HCF and write the answers inside the brackets

$$-ab - ac = -a(b + c)$$

$$-ab + ac = -a(b - c)$$

- To check that the factorised answer is correct, expand it

EXERCISE 1.06 ANSWERS ON P. 609

Factorising expressions U F

1 Find the HCF of each pair of terms.

EXAMPLE 10

a	$12, 6y$	**b**	$8a^2, 24$	**c**	$20b, 15$
d	$6, 9p^2$	**e**	$24mn, m$	**f**	c, c^2
g	$18pq, 12p^2$	**h**	$24w^2, 16w$	**i**	$3(x - 5), x(x - 5)$

2 Factorise each expression.

EXAMPLE 11

a	$16r^2 - 12r$	**b**	$6t^2 + 27t$	**c**	$36p^2 - 108p$
d	$12x^2y - 16x$	**e**	$18p^2 + 16pr$	**f**	$4m^2n - 4mn^2$
g	$14abc + 21bc$	**h**	$28vw - 21v^2w^2$	**i**	$45rt + 54r^2t$
j	$36pq^2r - 144pr$	**k**	$48x^2y + 64xy^2$	**l**	$75g^3h^2 - 125gh$
m	$25abc + 10a^2b$	**n**	$54my^3 + 45m^2y^2$	**o**	$21r^3u^2 - 35ru^3$

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3 Factorise completely $24xy^2 - 16xyw$. Select the correct answer **A**, **B**, **C** or **D**.

A $8xy(3y - 2w)$ **B** $2x(12y^2 - 8yw)$ **C** $4xy(6y - 4w)$ **D** $8xy^2(3 - 2w)$

4 Factorise each expression.

a $a(a - 3) + 6(a - 3)$ **b** $t(8 + t) - 3(8 + t)$ **c** $b(b + 5) - 2(b + 5)$

d $x(2 - y) - 6(2 - y)$ **e** $5(a - 7) + b(a - 7)$ **f** $y(a + 3) + 4(a + 3)$

g $3p(g + 5) - 2(g + 5)$ **h** $5(2 - 3m) + 4n(2 - 3m)$ **i** $r(8 + r) + (8 + r)$

j $(y - 6) - y(y - 6)$ **k** $3m(2w + 7) - (2w + 7)$ **l** $3c(c + 3) + 5(c + 3)$

EXAMPLE 12

5 Factorise each expression using the 'negative' HCF.

a $-4q - 8$ **b** $-9u + 18$ **c** $-3g + 6$ **d** $-18a + 12$

e $-n - 1$ **f** $-n + 1$ **g** $-y^2 - 9y$ **h** $-6t + 10t^2$

i $-3a^2 - 6ab$ **j** $-ap + aq$ **k** $-20e^2 - 22e$ **l** $-9m + 3m^2$

6 Factorise $-10kr + 4rn$. Select **A**, **B**, **C** or **D**.

A $-2r(5k - 4n)$ **B** $-2r(5k - 2n)$ **C** $-5r(2k - 2n)$ **D** $-2r(5k + 2n)$

7 Factorise each expression.

a $\frac{1}{2}p + \frac{1}{2}q$ **b** $7g + 9g^2$ **c** $\frac{1}{4}x^2 - \frac{1}{4}xy$

d $8a + 12y + 4$ **e** $20xy - 10x^2 + 5y$ **f** $t^2m + tm^2 + tm$

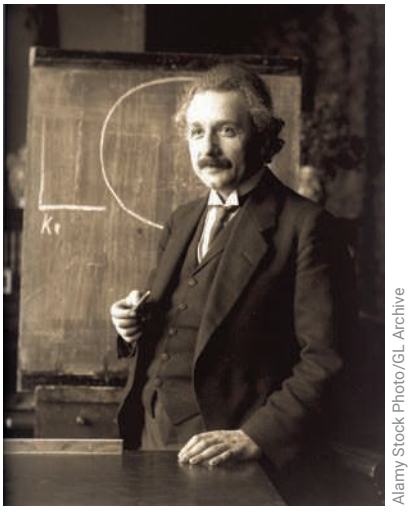
g $7y(m + 2) - p(m + 2)$ **h** $-10p^2 - 10pq$ **i** $5a(x - y) + 10a^2xy - 10a^2y^2$

Did you know?

Einstein's formula

In 1905, when the German scientist Albert Einstein was only 26, he proposed a new theory of physics for small particles of matter (such as atoms) moving at very high speeds. He proposed that matter (mass) could be converted into large amounts of energy, describing this in his Special Relativity formula $E = mc^2$, where E stands for energy, m stands for mass and c stands for the speed of light (3 billion metres per second). Einstein's theory revolutionised conventional laws of physics and led to the development of nuclear energy.

Find out why Einstein renounced his German citizenship and became an American in 1940.



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Expanding binomial products

1.07

$(x + 5)$ and $(x - 1)$ are called **binomial expressions** because they each have exactly 2 terms.

$(x + 5)(x - 1)$ is called a **binomial product** because it is a product of 2 binomial expressions.

binomial = '2 terms'

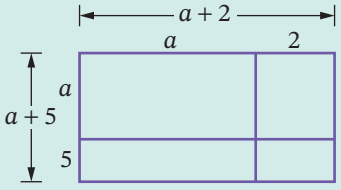
Example 13

Expand each binomial product using an area diagram.

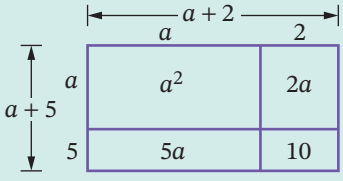
a $(a + 2)(a + 5)$ **b** $(n + 4)(n - 3)$

Solution

a Draw an area diagram (rectangle) with length $(a + 2)$ and width $(a + 5)$ and divide the diagram into 4 smaller rectangles.



Find the area of each smaller rectangle.

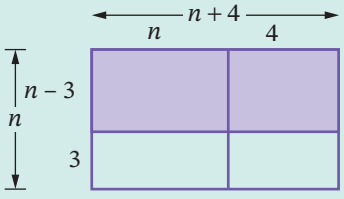


Expand $(a + 2)(a + 5)$ by adding the areas of the 4 rectangles.

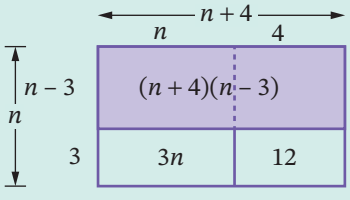
$(a + 2)(a + 5) = a^2 + 2a + 5a + 10$ Expanding

$= a^2 + 7a + 10$ Simplifying by collecting like terms

b Draw an area diagram with length $(n + 4)$ and width $(n - 3)$, then increase the width to n by adding 3. This creates 4 rectangles.



The 2 shaded upper rectangles together give the product $(n + 4)(n - 3)$. To find its area, we subtract the areas of the 2 lower rectangles from the area of the whole rectangle.



$(n + 4)(n - 3) = n(n + 4) - 3n - 12$ Whole rectangle − 2 lower rectangles

$= n^2 + 4n - 3n - 12$ Expanding

$= n^2 + n - 12$ Simplifying by collecting like terms

When expanding a binomial product algebraically, each term in the first binomial is multiplied by each term in the second binomial to give **4 terms**, which are collected and simplified.

Video
Expanding binomial products 1

Example 14

Expand each binomial product.

a $(x + 5)(x + 9)$ **b** $(k + 3)(k - 7)$ **c** $(7 - m)(4 + m)$ **d** $(a - 6)^2$

Solution

a $(x + 5)(x + 9) = x(x + 9) + 5(x + 9)$ — Each term in $(x + 5)$ is multiplied by $(x + 9)$
$= x^2 + 9x + 5x + 45$ — Expanding to make 4 terms
$= x^2 + 14x + 45$ — Adding $9x$ and $5x$

b $(k + 3)(k - 7) = k(k - 7) + 3(k - 7)$ — Each term in $(k + 3)$ is multiplied by $(k - 7)$
$= k^2 - 7k + 3k - 21$ — Expanding to make 4 terms
$= k^2 - 4k - 21$ — Adding $-7k$ and $3k$

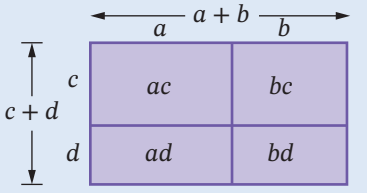
c $(7 - m)(4 + m) = 7(4 + m) - m(4 + m)$ — Each term in $(7 - m)$ is multiplied by $(4 + m)$
$= 28 + 7m - 4m - m^2$ — Expanding to make 4 terms
$= 28 + 3m - m^2$ — Adding $7m$ and $-4m$

d $(a - 6)^2 = (a - 6)(a - 6)$
$= a(a - 6) - 6(a - 6)$ — Each term in $(a - 6)$ is multiplied by $(a - 6)$
$= a^2 - 6a - 6a + 36$ — Expanding to make 4 terms
$= a^2 - 12a + 36$ — Adding $-6a$ and $-6a$

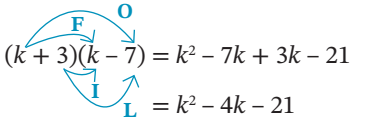
The distributive law for expanding a binomial product

Multiply each term in the first binomial by each term in the second binomial.

$$(a + b)(c + d) = ac + ad + bc + bd$$



One way of remembering which pairs of terms to multiply together in a binomial product is called the **FOIL method**, as shown below.



- **F** means multiply the **first** terms: $k \times k = k^2$
- **O** means multiply the **outside** terms: $k \times (-7) = -7k$
- **I** means multiply the **inside** terms: $3 \times k = 3k$
- **L** means multiply the **last** terms: $3 \times (-7) = -21$

Example 15

Expand each binomial product.

a $(x - 6)(4x + 2)$ **b** $(3t - 1)(2t - 5)$ **c** $(2x + 5)(3y - 4)$

Solution

a $(x - 6)(4x + 2) = x(4x + 2) - 6(4x + 2)$
$= 4x^2 + 2x - 24x - 12$ Expanding
$= 4x^2 - 22x - 12$ Simplifying

b $(3t - 1)(2t - 5) = 3t(2t - 5) - 1(2t - 5)$
$= 6t^2 - 15t - 2t + 5$ Expanding
$= 6t^2 - 17t + 5$ Simplifying

c $(2x + 5)(3y - 4) = 2x(3y - 4) + 5(3y - 4)$
$= 6xy - 8x + 15y - 20$ Expanding

Videos
Expanding binomial products 1
Expanding binomial products 2

1.07

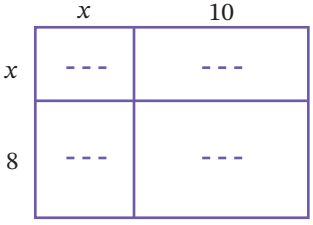
EXAMPLE 13

EXERCISE 1.07 ANSWERS ON P. 609

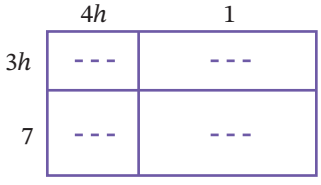
Expanding binomial products UFRC

1 Expand each binomial product by copying and completing the area diagram.

a $(x + 10)(x + 8)$

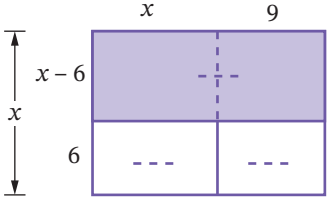


b $(4h + 1)(3h + 7)$

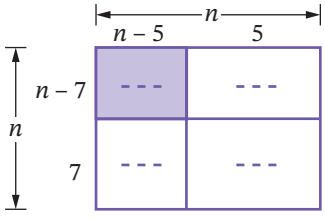


2 Expand each binomial product by copying and completing the area diagram. (Find the shaded area.)

a $(x + 9)(x - 6)$



b $(n - 5)(n - 7)$



3 Draw an area diagram for each binomial product and use it to expand the product.

a $(2y + 5)(y + 3)$ **b** $(w - 5)(w + 3)$ **c** $(p - 3)(p - 5)$

Foundation Standard Complex

EXAMPLE 14

4 Expand each binomial product algebraically.

a	$(n + 6)(n + 3)$	**b**	$(t + 6)(t + 5)$	**c**	$(p + 10)(p - 10)$
d	$(x - 8)(x + 3)$	**e**	$(b - 2)(9 + b)$	**f**	$(u - 8)(u - 7)$
g	$(15 - r)(r + 1)$	**h**	$(a - 10)(a - 9)$	**i**	$(5 - c)(3 - c)$
j	$(t - 1)(t + 2)$	**k**	$(y - 4)(y + 10)$	**l**	$(n - 9)(11 + n)$
m	$(e - 2)^2$	**n**	$(5 + w)^2$	**o**	$(g + 11)^2$

5 Expand $(b + 7)^2$. Select the correct answer **A**, **B**, **C** or **D**.

A $b^2 + 49$ **B** $b^2 + 49b$ **C** $b^2 + 7b + 49$ **D** $b^2 + 14b + 49$

6 Expand $(x + 8)(3 - x)$. Select **A**, **B**, **C** or **D**.

A $24 - 5x - x^2$ **B** $-x^2 + 24$ **C** $x^2 + 5x - 24$ **D** $-x^2 + 5x + 24$

EXAMPLE 15

7 Expand each binomial product.

a	$(2x + 5)(x + 3)$	**b**	$(3e + 2)(4e + 5)$	**c**	$(10 + 3p)(p - 1)$
d	$(7d - 2)(7d - 2)$	**e**	$(2f - 2)(3f + 5)$	**f**	$(4m - 5)(5 + 3m)$
g	$(3 - 4h)(2 + 5h)$	**h**	$(4p - 5)(2p - 7)$	**i**	$(2m - 3)(4 - 5m)$
j	$(3t + 5)(2t - 1)$	**k**	$(5y - 5)(5y + 5)$	**l**	$(7 - 2a)(2a - 7)$
m	$(3k + 8)^2$	**n**	$(4c - 5)^2$	**o**	$(2d + 9)(3x - 1)$
p	$(5 - 2g)(h + 3)$	**q**	$(7 - 4w)^2$	**r**	$(3n - 5)(4n + 9)$

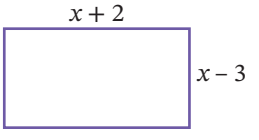
8 Expand $(3h + 8)(2h - 5)$. Select **A**, **B**, **C** or **D**.

A $6h^2 - 31h - 40$ **B** $6h^2 - h + 40$

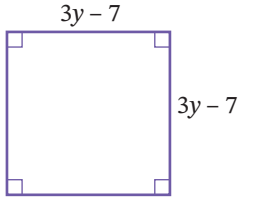
C $6h^2 + 31h - 40$ **D** $6h^2 + h - 40$

9 Write a simplified algebraic expression for the area of each shape.

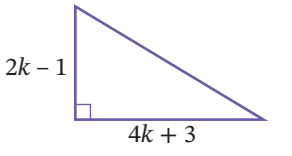
a



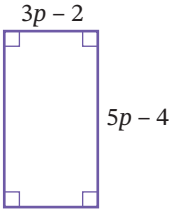
b



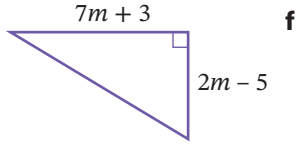
c



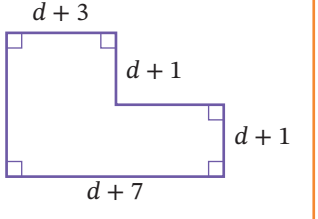
d



e



f



Foundation Standard Complex

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10 A rectangular barbecue plate has a length of 100 cm and a width of 75 cm. The length and width are both increased by x cm. R C

a Write an expression for the new length of the plate in cm.

b Write an expression for the new width of the plate in cm.

c Hence find a simplified expression for the new area of the plate in cm^2.

d By how much has the area of the plate increased?

e If $x = 0.1$, find the increase in the area of the plate.

11 A family room in a house is to be extended. The room is 4 m long and 3 m wide. The length is to be increased by x m and the width by y m. R C

a Write down expressions for the new length and width in metres.

b Write down a binomial expression for the new area of the room in square metres.

c Expand and simplify your expression for the area.

d By how much has the area of the room increased?

12 Prove that: R C

a $(a - b)^2 = (b - a)^2$ **b** $(a + b)(a - b) = a^2 - b^2$ **c** $(a - b)^2 = a^2 - 2ab + b^2$

Investigation

Expanding perfect squares

There is a special pattern when you expand a binomial by itself, for example, $(y + 5)(y + 5)$. This is called the square of a binomial, or a perfect square.

1 **a** Expand and simplify the perfect square $(y + 5)^2 = (y + 5)(y + 5)$.

b How many terms are there in your answer?

c The terms in the binomial $(y + 5)$ are y and 5. The first term of your *answer* in part **a** is y^2, which is the square of y. How are the other terms in the answer related to y and 5?

2 Expand and simplify each perfect square.

a $(k + 3)^2$ **b** $(m + 7)^2$ **c** $(p + 2)^2$

3 For each of the expansions in question **2**:

a how many terms are there?

b describe how each is related to the 2 terms of the perfect square

c compare and discuss your results with other students

4 Expand and simplify each perfect square.

a $(t - 1)^2$ **b** $(g - 6)^2$ **c** $(d - 5)^2$

5 For each of the expansions in question **4**:

a how many terms are there?

b describe how each is related to the 2 terms of the perfect square

c compare and discuss your results with other students

Foundation Standard Complex

6 Describe the expansion of the perfect square (first term + second term)^2 by copying and completing this statement:

The square of a binomial is equal to the square of the first term plus double the product of the 2 terms plus _____

7 **a** Find a formula for $(a + b)^2$.

b Check this formula by expanding $(a + b)^2$ using an area diagram.

c Find a formula for $(a - b)^2$.

d Check this formula by expanding $(a - b)^2$ using an area diagram.

1.08 Perfect squares

PATH

Puzzles
Factorominoes

Perfect squares

16, 49, v^2 and $(y + 5)^2$ are called **perfect squares** because they are square numbers.

Perfect squares

The formulas for expanding the perfect square of a binomial are:

$$(a + b)^2 = a^2 + 2ab + b^2$$
$$(a - b)^2 = a^2 - 2ab + b^2$$

Proof:

$$\begin{aligned}(a + b)^2 &= (a + b)(a + b)\\ &= a(a + b) + b(a + b)\\ &= a^2 + ab + ba + b^2\\ &= a^2 + 2ab + b^2\end{aligned}$$

$$\begin{aligned}(a - b)^2 &= (a - b)(a - b)\\ &= a(a - b) - b(a - b)\\ &= a^2 - ab - ba + b^2\\ &= a^2 - 2ab + b^2\end{aligned}$$

Example 16

Copy and complete the expansion of each perfect square.

a $(x + 4)^2 = x^2 + ____ + 16$

b $(y - 6)^2 = y^2 - 12y + ____$

c $(5g + 9)^2 = ____ + 90g + 81$

d $(3d - 5)^2 = 9d^2 - ____ + 25$

Solution

a In the expansion,

$_____ = 2 \times x \times 4$ Doubling the product of the 2 terms

$= 8x$

$\therefore (x + 4)^2 = x^2 + 8x + 16$

b In the expansion,

$_____ = 6^2$ The second term squared

$= 36$

$\therefore (y - 6)^2 = y^2 - 12y + 36$

c $_____ = (5g)^2$ — The first term squared
$= 25g^2$
$\therefore (5g + 9)^2 = 25g^2 + 90g + 81$

d $_____ = 2 \times 3d \times 5$ — Doubling the product of the 2 terms
$= 30d$
$\therefore (3d - 5)^2 = 9d^2 - 30d + 25$

Example 17

Expand each perfect square.

a $(n - 5)^2$ **b** $(k + 4)^2$ **c** $(3y - 8)^2$

Solution

a $(n - 5)^2 = n^2 - 2 \times n \times 5 + 5^2$ — 1st term squared – double product + 2nd term squared
$= n^2 - 10n + 25$

b $(k + 4)^2 = k^2 + 2 \times k \times 4 + 4^2$
$= k^2 + 8k + 16$

c $(3y - 8)^2 = (3y)^2 - 2 \times 3y \times 8 + 8^2$
$= 9y^2 - 48y + 64$

EXERCISE 1.08 ANSWERS ON P. 610

Perfect squares UFRC

1 Copy and complete the expansion of each perfect square.

a $(x + 10)^2 = x^2 + _____ + 100$
b $(m - 8)^2 = _____ - 16m + 64$
c $(p - t)^2 = p^2 - 2pt + _____$
d $(h + 4)^2 = h^2 _____ + 16$
e $(k - 9)^2 = k^2 _____ + 81$
f $(8 + 5f)^2 = 64_____ + 25f^2$
g $(2d + 3)^2 = _____ + _____ + 9$
h $(6a + 1)^2 = _____ + 12a + _____$

2 Expand each perfect square.

a $(m + 9)^2$ **b** $(u + 3)^2$ **c** $(y - 6)^2$ **d** $(8 + k)^2$
e $(5 - h)^2$ **f** $(7 + k)^2$ **g** $(f + 20)^2$ **h** $(q - 11)^2$
i $(10 + t)^2$ **j** $(x - w)^2$ **k** $(a + g)^2$ **l** $(2m - 3)^2$
m $(5x - 6)^2$ **n** $(9a + 2)^2$ **o** $(3e - 4)^2$ **p** $(5 + 7b)^2$
q $(4 - 5p)^2$ **r** $(11 - 2c)^2$ **s** $(10g + 3)^2$ **t** $(3k + 11)^2$

3 Expand each perfect square.

a $(7h + 2k)^2$ **b** $(8a - 3y)^2$ **c** $(xy + z)^2$
d $\left(1 + \frac{1}{y}\right)^2$ **e** $\left(k - \frac{1}{k}\right)^2$ **f** $\left(w + \frac{3}{w}\right)^2$

PATH
1.08
Video Special binomial products
EXAMPLE 16
EXAMPLE 17

Foundation Standard Complex

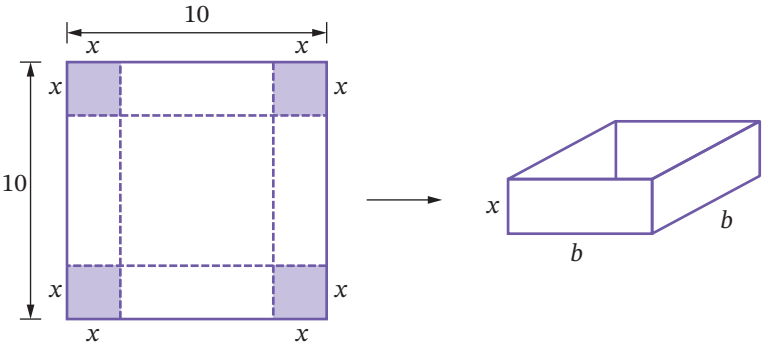
PATH

4 Use the expansion of a perfect square to evaluate each square number without using a calculator.

a $21^2 = (20 + 1)^2$ **b** $45^2 = (40 + 5)^2$ **c** $29^2 = (30 - 1)^2$

d $59^2 = (60 - 1)^2$ **e** $102^2 = (100 + 2)^2$ **f** $98^2 = (100 - 2)^2$

5 A 10 cm square sheet of cardboard has a square of dimension x cm cut from each corner. It is then folded to form a square-based prism. R C



a Why is the length of the square base (b) equal to $(10 - 2x)$ cm?

b Find the area of the square base.

c Find the area of one side-face of the prism.

d Hence show that the surface area of the prism is $(100 - 4x^2)$ cm^2.

Investigation

Squaring a number ending in 5

Study this mental shortcut for squaring a number ending in 5:

- to evaluate 35^2, calculate $3 \times 4 = 12$, append '25' to the end: $35^2 = 1225$
- to evaluate 75^2, calculate $7 \times 8 = 56$, append '25' to the end: $75^2 = 5625$
- to evaluate 105^2, calculate $10 \times 11 = 110$, append '25' to the end: $105^2 = 11\,025$

Let n stand for the tens digit of the number ending in 5 being squared.

Expand $(10n + 5)^2$ and investigate why the above method works.

Investigation

Expanding sums by differences

There is also special pattern when you multiply the sum of 2 terms by the difference of those 2 terms, for example, $(x + 9)(x - 9)$.

1 **a** Expand and simplify $(x + 9)(x - 9)$.

b How many terms are there in your answer?

c How are the terms in the answer related to x and 9?

Foundation Standard Complex

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2 Expand and simplify each 'sum by difference'.

a $(k + 2)(k - 2)$ **b** $(m + 5)(m - 5)$ **c** $(p - 6)(p + 6)$

3 Describe the expansion of (first term + second term)(first term – second term) by copying and completing this statement:

The product of a 'sum by difference' *is equal to the square of the first term* _____

_____

4 **a** Find a formula for $(a + b)(a - b)$.

b Check this formula by expanding $(a + b)(a - b)$ using an area diagram.

PATH

Difference of 2 squares 1.09

Difference of 2 squares

$$(a + b)(a - b) = a^2 - b^2$$

The answer is called the **difference of 2 squares**.

Proof:

$$\begin{aligned}(a - b)(a + b) &= a(a + b) - b(a + b)\\ &= a^2 + ab - ba - b^2\\ &= a^2 - b^2\end{aligned}$$

When the sum of 2 terms is multiplied by their difference, the answer is the square of the first term minus the square of the second term (the difference of 2 squares).

PATH

Worksheet
Special products

Puzzle
Enough time

Videos
Difference of 2 squares

Special binomial products

Example 18

Expand each expression.

a $(d + 3)(d - 3)$ **b** $(2 + r)(2 - r)$

c $(7x + 2)(7x - 2)$ **d** $(4k - 5p)(4k + 5p)$

Solution

a $(d + 3)(d - 3) = d^2 - 3^2$
$= d^2 - 9$

b $(2 + r)(2 - r) = 2^2 - r^2$
$= 4 - r^2$

c $(7x + 2)(7x - 2) = (7x)^2 - 2^2$
$= 49x^2 - 4$

d $(4k - 5p)(4k + 5p) = (4k)^2 - (5p)^2$
$= 16k^2 - 25p^2$

PATH

EXERCISE 1.09 ANSWERS ON P. 610

Difference of 2 squares UFRC

EXAMPLE 18

1 Expand each expression.

a	$(m+5)(m-5)$	**b**	$(c-10)(c+10)$	**c**	$(a+12)(a-12)$
d	$(6-y)(6+y)$	**e**	$(8-m)(8+m)$	**f**	$(p+1)(p-1)$
g	$(5+e)(5-e)$	**h**	$(v+11)(v-11)$	**i**	$(w-3)(w+3)$
j	$(x-10)(x+10)$	**k**	$(q+7)(q-7)$	**l**	$(9-g)(9+g)$
m	$(b-2)(b+2)$	**n**	$(15-r)(15+r)$	**o**	$(d+13)(d-13)$

2 Expand each expression.

a	$(2h-3)(2h+3)$	**b**	$(5r+4)(5r-4)$	**c**	$(5b+8)(5b-8)$
d	$(4p-7)(4p+7)$	**e**	$(3-8k)(3+8k)$	**f**	$(7x-5)(7x+5)$
g	$(2+9m)(2-9m)$	**h**	$(9k-4l)(9k+4l)$	**i**	$(7n+8m)(7n-8m)$
j	$(4g-5h)(4g+5h)$	**k**	$(7u+3w)(7u-3w)$	**l**	$(11a+3b)(11a-3b)$
m	$\left(a+\frac{1}{a}\right)\left(a-\frac{1}{a}\right)$	**n**	$\left(\frac{w}{3}-2\right)\left(\frac{w}{3}+2\right)$	**o**	$\left(1-\frac{1}{r}\right)\left(1+\frac{1}{r}\right)$

3 Natalie's age is p years. R C

a What was Natalie's age last year?

b What will Natalie's age be next year?

c Write an expression for (Natalie's age last year) × (Natalie's age next year).

d If (Natalie's age last year) × (Natalie's age next year) is equal to 48, what is Natalie's age?

4 By expressing 31×29 as $(30+1)(30-1)$, use the difference of 2 squares to find the value of 31×29. R C

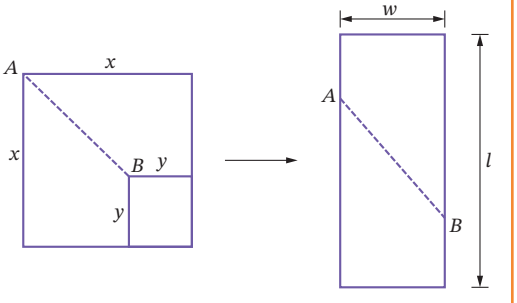
5 Use the method of question 4 to evaluate each expression. R C

a 21×19 **b** 51×49 **c** 89×91 **d** 78×82

6 Use 'sum by difference' to evaluate each expression.

a	$15^2 - 14^2$	**b**	$24^2 - 23^2$	**c**	$65^2 - 35^2$
d	$101^2 - 99^2$	**e**	$23^2 - 17^2$	**f**	$50^2 - 48^2$

7 A square sheet of paper of length x cm has a smaller square of length y cm cut from one corner. It is then cut on the diagonal AB and rearranged to form a rectangle as shown. R C



a Find an expression for:

i the length (l) of the rectangle

ii the width (w) of the rectangle

iii the area of the rectangle (do not simplify)

b The area of the rectangle should be the same as the area of the square on the left minus the area of the smaller square. Find an expression for this area. What does this prove?

Foundation Standard Complex

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Mixed expansions 1.10

Example 19

Expand and simplify each expression.

a $(4r + 5)(1 - 2r)$ **b** $(7 + 9x)^2$

c $(3d - 10)(3d + 10)$ **d** $(a + 6)(a - 6) + (a + 12)(a + 3)$

e $(m - 2)^2 - (m - 2)(m + 2)$

Solution

a $(4r + 5)(1 - 2r) = 4r(1 - 2r) + 5(1 - 2r)$
$= 4r - 8r^2 + 5 - 10r$
$= -8r^2 - 6r + 5$

b $(7 + 9x)^2 = 7^2 + 2 \times 7 \times 9x + (9x)^2$
$= 49 + 126x + 81x^2$
$= 81x^2 + 126x + 49$

c $(3d - 10)(3d + 10) = (3d)^2 - 10^2$
$= 9d^2 - 100$

d $(a + 6)(a - 6) + (a + 12)(a + 3) = a^2 - 36 + a^2 + 3a + 12a + 36$
$= 2a^2 + 15a$

e $(m - 2)^2 - (m - 2)(m + 2) = m^2 - 4m + 4 - (m^2 - 4)$
$= m^2 - 4m + 4 - m^2 + 4$
$= -4m + 8$

EXERCISE 1.10 ANSWERS ON P. 610

Mixed expansions U F R C

1 Expand and simplify each expression.

a $(2m - 1)(2m + 1)$ **b** $(y + 4)(5y - 3)$ **c** $(2k - 7)^2$

d $(d + 9)^2$ **e** $(2e - 1)(e + 1)$ **f** $(5a + 4)(5a - 4)$

g $(2 - p)(p - 2)$ **h** $(10 - 6y)(10 + 6y)$ **i** $(h - 3m)^2$

j $(2x - 3)(y + 3)$ **k** $(11a - 4b)(11a + 4b)$ **l** $\left(u - \frac{1}{u}\right)^2$

2 Expand and simplify each expression. R C

a $(m - 5)(m + 5) + 25$ **b** $6y + (y - 3)^2 + 9$

c $(3x + 1)(2 - x) + 2x + 4$ **d** $(d + 4)^2 - 8d + 5$

e $16 + (4k - 8)(4k + 8)$ **f** $(x - y)^2 - (x + y)^2$

g $20a - (a - 2)(a - 5) + a^2$ **h** $2(f - 2)(f + 2)$

i $(2h + 3)^2 - (2h - 3)(2h + 3)$ **j** $7xy - (2x - 3)(y + 3)$

PATH

1.10

Puzzle
Expanding binomials

Worksheets
Algebra 6
Mixed expansions

EXAMPLE 19

Foundation Standard Complex

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PATH

3 Expand and simplify each expression. R C

a $(8a-1)(8a+1)-4a^2+1$ **b** $(n+1)^2+2n+3$

c $3(4-u)(4+u)+(u-12)(u+4)$ **d** $(2m-n)^2+(2m+n)^2$

e $(x-2)(x+3)-(x-2)(x+2)$ **f** $2(b-1)^2-(2b-1)^2$

g $(y+1)^2+(y+2)^2+(y+3)^2$ **h** $(x-3)(x+3)+(x+3)^2+(x-3)^2$

i $(5n+3)(5n-3)+(3n-5)(3n+5)$ **j** $2(a-b)(a+b)-(a+b)^2-(a-b)^2$

Quiz
Mental skills 1B

Mental skills 1B: Maths without calculators ANSWERS ON P. 610

Multiplying by 9, 11, 99 and 101

We can use expansion when multiplying by a number near 10 or near 100.

1 Study each example.

a
$$\begin{aligned}25\times11&=25\times(10+1)\\&=25\times10+25\times1\\&=250+25\\&=275\end{aligned}$$

b
$$\begin{aligned}14\times9&=14\times(10-1)\\&=14\times10-14\times1\\&=140-14\\&=126\end{aligned}$$

c
$$\begin{aligned}32\times12&=32\times(10+2)\\&=32\times10+32\times2\\&=320+64\\&=384\end{aligned}$$

d
$$\begin{aligned}7\times99&=7\times(100-1)\\&=7\times100-7\times1\\&=700-7\\&=693\end{aligned}$$

e
$$\begin{aligned}27\times101&=27\times(100+1)\\&=27\times100+27\times1\\&=2700+27\\&=2727\end{aligned}$$

f
$$\begin{aligned}18\times8&=18\times(10-2)\\&=18\times10-18\times2\\&=180-36\\&=144\end{aligned}$$

2 Now evaluate each product.

a	16×11	**b**	33×11	**c**	29×9	**d**	45×9
e	62×11	**f**	7×101	**g**	18×101	**h**	36×99
i	19×8	**j**	45×12	**k**	21×102	**l**	6×98
m	32×9	**m**	7×99	**o**	39×101	**p**	71×12

Foundation Standard Complex

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Factorising special binomial products 1.11

Factorising by grouping in pairs

An algebraic expression with 4 terms can often be factorised in pairs, that is, 2 terms at a time, to make a binomial product.

Example 20

Factorise each expression.

a $3ac + 2bd + 2bc + 3ad$ **b** $4km + 6mn - 6kp - 9np$

c $10xw - 6yw - 10xt + 6yt$

Solution

a $3ac + 2bd + 2bc + 3ad = 3ac + 3ad + 2bd + 2bc$ Grouping into pairs for factorising

$= 3a(c + d) + 2b(d + c)$ Factorising each pair

$= (c + d)(3a + 2b)$ Factorising again

b $4km + 6mn - 6kp - 9np = 2m(2k + 3n) - 3p(2k + 3n)$ Factorising each pair

$= (2k + 3n)(2m - 3p)$ Factorising again

c $10xw - 6yw - 10xt + 6yt = 2(5xw - 3yw - 5xt + 3yt)$ Factorising all terms first

$= 2(5xw - 5xt - 3yw + 3yt)$ Grouping into pairs for factorising

$= 2[5x(w - t) - 3y(w - t)]$ Factorising each pair

$= 2(w - t)(5x - 3y)$ Factorising again

Factorising the difference of 2 squares

We have learned the formula $(a + b)(a - b) = a^2 - b^2$.

If we use this rule in reverse, then the factors of $a^2 - b^2$ are $(a + b)$ and $(a - b)$.

Factorising the difference of 2 squares

$$a^2 - b^2 = (a + b)(a - b)$$

Example 21

Factorise each expression.

a $x^2 - 4$ **b** $9 - 16b^2$ **c** $20d^2 - 5a^2$ **d** $y^3 - y$

Solution

a $x^2 - 4 = x^2 - 2^2$

$= (x + 2)(x - 2)$

b $9 - 16b^2 = 3^2 - (4b)^2$

$= (3 + 4b)(3 - 4b)$

c $20d^2 - 5a^2 = 5(4d^2 - a^2)$

$= 5[(2d)^2 - a^2]$

$= 5(2d + a)(2d - a)$

d $y^3 - y = y(y^2 - 1)$

$= y(y + 1)(y - 1)$

PATH

1.11

Puzzle
Grouping

Video
Factorising by grouping in pairs

Puzzle
Difference of 2 perfect squares

Video
Difference of 2 squares

PATH

EXERCISE 1.11 ANSWERS ON P. 610

Factorising special binomial products U F

EXAMPLE 20

1 Factorise each expression.

a $4ab + 5bc + 4ad + 5cd$
b $2xy - 5wy + 2xt - 5wt$
c $9ac + 6bc + 12ad + 8bd$
d $10x^2 + 30 + x^3 + 3x$
e $3a^2 + 3ab + 3ac + 3bc$
f $6rt - 18wt + 6rp - 18wp$
g $14e - 21 + 2de - 3d$
h $hk - h^2 - 2k + 2h$
i $3mn - 6m + pn - 2p$
j $9p^2 - 27 + qp^2 - 3q$
k $fg - fh - 10g + 10h$
l $9kl - 12ml + 9kn - 12mn$
m $2p - 2c - p^2 + pc$
n $l^3 + lm^2 - 3l^2 - 3m^2$
o $a(x + 1) + y(x + 1) - ka - ky$
p $p(a - b) - 2q(a - b) + 3qp - 6q^2$

EXAMPLE 21

2 Factorise each expression.

a $w^2 - 9$
b $y^2 - 36$
c $k^2 - 1$
d $m^2 - 121$
e $p^2 - 64$
f $c^2 - 100$
g $4e^2 - f^2$
h $a^2 - 9b^2$
i $16y^2 - 1$
j $4 - b^2$
k $25 - e^2$
l $1 - 16x^2$
m $k^2 - u^2$
n $49 - 16m^2$
o $b^2 - 121d^2$
p $36c^2 - 25k^2$
q $16 - 81h^2$
r $25a^2 - 64m^2$
s $100 - 49n^2$
t $121p^2 - 144q^2$
u $\frac{1}{4} - 25c^2$
v $4w^2 - \frac{1}{9}$
w $64h^2 - 2\frac{1}{4}$
x $1 - m^2n^2$

3 Factorise each expression.

a $2a^2 - 2b^2$
b $7k^2 - 28$
c $3 - 75u^2$
d $x^3 - 49x$
e $k - 16k^3$
f $50q^2 - 2$
g $3d^2 - 12v^2$
h $5t^5 - 125t^3$
i $2a^2b^2 - 2$
j $x^2y^2 - x^2w^2$
k $192f^2 - 108g^2$
l $45d^2 - \frac{5}{4}$
m $2x^2 - 8a^2$
n $100 - 25w^2$
o $1\frac{1}{4} - 80e^2$
p $9c^2 - 6\frac{1}{4}$

4 Factorise each expression.

a $\frac{p^2}{16} - \frac{b^2}{25}$
b $x^2 - \frac{1}{9}$
c $\frac{v^2}{64} - \frac{u^2}{81}$
d $\frac{2y^2}{9} - \frac{2m^2}{121}$
e $\frac{16a^2}{49} - \frac{25b^2}{4}$
f $g^4 - 81$
g $100 - n^4$
h $(x + y)^2 - x^2$
i $(p - 2q)^2 - (2p + q)^2$
j $\frac{x^2}{4} - \frac{y^2}{36}$
k $(a + b)^2 - (a - b)^2$
l $25 - (4 - x)^2$

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Investigation

Factorising quadratic expressions

1 **a** Show that $(x + 3)(x + 5) = x^2 + 8x + 15$.

b The quadratic expression $x^2 + 8x + 15$ has 3 terms. The coefficient of x is 8, the number in front of the x. How are the 3 and 5 in $(x + 3)(x + 5)$ related to the 8?

c The constant term in $x^2 + 8x + 15$ is 15, the number with no x at the end. How are the 3 and 5 related to the 15?

2 **a** Expand $(x + 9)(x + 2)$.

b What is the coefficient of x? How are 9 and 2 related to it?

c What is the constant term? How are 9 and 2 related to it?

3 **a** Expand $(x + 8)(x - 3)$.

b What is the coefficient of x? How are 8 and -3 related to it?

c What is the negative constant term? How are 8 and -3 related to it?

4 **a** Expand $(x - 4)(x - 1)$.

b What is the negative coefficient of x? How are -4 and -1 related to it?

c What is the positive constant term? How are -4 and -1 related to it?

5 In the expansion of any binomial product, how are the coefficient of x and the constant term related to the numbers in the binomials?

6 Copy and complete:

a $(x + ___)(x + ___) = x^2 + 5x + 4$ **b** $(x + ___)(x + ___) = x^2 + 8x + 15$

c $(x + ___)(x + ___) = x^2 + 7x + 12$ **d** $(x + ___)(x - ___) = x^2 - 4x - 32$

e $(x + ___)(x - ___) = x^2 + 2x - 3$ **f** $(x - ___)(x - ___) = x^2 - 9x + 20$

PATH

1.12

Factorising quadratic expressions $x^2 + bx + c$

1.12

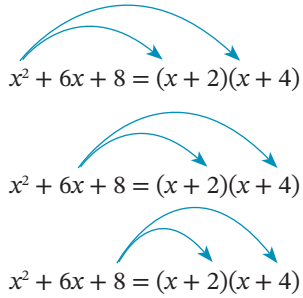
A quadratic expression such as $x^2 - 5x + 7$ is called a **trinomial** because it has 3 terms. Other examples of quadratic trinomials are $x^2 + x - 15$, $2x^2 - 3x + 9$ and $-4x^2 + 9x + 20$.

The expansion of $(x + 2)(x + 4)$ is $x^2 + 6x + 8$.

$\therefore$ The factorisation of $x^2 + 6x + 8$ is $(x + 2)(x + 4)$.

When factorising a quadratic trinomial such as $x^2 + 6x + 8$:

- each factor must have an x term to give x^2
- $2 + 4 = 6$, which is the **coefficient** of x, the number in front of the x
- $2 \times 4 = 8$, which is the **constant term** with no x.



PATH

Puzzle
Perfect squares

Technology
Factorising trinomials

Spreadsheeet
Factorising trinomials

PATH

Video
Factorising quadratic expressions 1

Example 22

Factorise each quadratic trinomial.

a $a^2 + 7a + 12$ **b** $x^2 + x - 6$

Solution

a Find the 2 numbers that have a sum of 7 and a product of 12.
It is best to test numbers that have a product of 12 and then check if their sum equals 7:
The correct numbers are 3 and 4.
$\therefore a^2 + 7a + 12 = (a + 3)(a + 4)$

Pair of numbers	Product	Sum
6, 2	6 × 2 = 12	6 + 2 = 8
3, 4	3 × 4 = 12	3 + 4 = 7

b Find 2 numbers with a sum of 1 and a product of –6.
Test numbers that have a product of –6 and check their sums:
The correct numbers are 3 and –2.
$\therefore x^2 + x - 6 = (x + 3)(x - 2)$

Pair of numbers	Product	Sum
+6, −1	+6 × (−1) = −6	+6 + (−1) = 5
+2, −3	+2 × (−3) = −6	2 + (−3) = −1
+3, −2	+3 × (−2) = −6	+3 + (−2) = 1

Factorising quadratic trinomials of the form $x^2 + bx + c$

- Find 2 numbers that have a sum of b and a product of c
- Use these 2 numbers to write a binomial product of the form $(x ___)(x ___)$

Video
Factorising quadratic expressions 1

Example 23

Factorise each quadratic expression.

a $a^2 - 2a - 15$ **b** $y^2 - 6y + 8$ **c** $x^2 - 2x + 1$

Solution

a $a^2 - 2a - 15$
Find 2 numbers that have a product of –15 and a sum of –2. Since the product is negative, one of the numbers must be negative. They are –5 and +3.
$\therefore a^2 - 2a - 15 = (a - 5)(a + 3)$

b $y^2 - 6y + 8$
Product = 8, sum = −6.
Since the sum is negative, one of the numbers must be negative, but since the product is positive, *both* of the numbers must be negative. They are −2 and −4.
$\therefore y^2 - 6y + 8 = (y - 2)(y - 4)$

c $x^2 - 2x + 1$
Product = 1, sum = −2. Since the sum is negative, one of the numbers must be negative, but since the product is positive, *both* of the numbers must be negative.
They are −1 and −1.
$\therefore x^2 - 2x + 1 = (x - 1)(x - 1)$
$= (x - 1)^2$ *Note:* $(x - 1)^2$ is a perfect square.

Example 24

Factorise each quadratic trinomial.

a $3g^2 + 12g - 36$ **b** $48 - 8p - p^2$

Solution

a $3g^2 + 12g - 36 = 3(g^2 + 4g - 12)$ Taking out the HCF of 3 first
$= 3(g - 2)(g + 6)$ Product = –12, sum = 4

b $48 - 8p - p^2 = -p^2 - 8p + 48$ Rearranging the terms to make the p^2 term first
$= -1(p^2 + 8p - 48)$ Taking out a common factor of –1
$= -(p + 12)(p - 4)$ Product = –48, sum = 8

EXERCISE 1.12 ANSWERS ON P. 611

Factorising quadratic expressions $x^2 + bx + c$ U F R

1 Find 2 numbers whose: R

a product is 6 and their sum is −7
b product is −12 and their sum is 1
c product is −15 and their sum is −2
d product is 12 and their sum is 7
e product is 20 and their sum is −9
f product is −14 and their sum is 5
g product is −10 and their sum is 3
h product is−25 and their sum is 0
i product is −2 and their sum is −1
j product is −18 and their sum is −7

2 Factorise each quadratic expression. R

EXAMPLE 22

a $m^2 + 7m + 12$ **b** $x^2 + 12x + 35$ **c** $k^2 + 5k + 4$
d $p^2 + 7p + 10$ **e** $w^2 + 9w + 20$ **f** $b^2 + 6b + 5$
g $e^2 + 5e + 6$ **h** $r^2 + 4r + 4$ **i** $n^2 + 11n + 10$
j $a^2 + 11a + 30$ **k** $d^2 + 10d + 24$ **l** $y^2 + 15y + 44$

3 Factorise each quadratic expression. R

a $p^2 - 5p + 4$ **b** $v^2 - 11v + 30$ **c** $y^2 - 7y + 12$
d $x^2 - 12x + 20$ **e** $a^2 - 9a + 18$ **f** $e^2 - 8e + 12$
g $c^2 - 14c + 48$ **h** $w^2 - 5w + 6$ **i** $d^2 - 18d + 77$
j $u^2 - 3u + 2$ **k** $n^2 - 8n + 16$ **l** $b^2 - 15b + 56$

4 Factorise each quadratic expression. R

EXAMPLE 23

a $x^2 + 3x - 4$ **b** $g^2 + 5g - 24$ **c** $y^2 - 2y - 3$
d $r^2 - 5r - 14$ **e** $m^2 + 2m - 15$ **f** $a^2 + a - 2$
g $m^2 - 9m + 20$ **h** $h^2 - 3h - 4$ **i** $d^2 - 6d + 9$
j $w^2 + 4w - 12$ **k** $a^2 - 4a - 12$ **l** $k^2 + 5k - 14$
m $c^2 - 7c - 18$ **n** $e^2 - 6e - 27$ **o** $u^2 + 6u - 55$
p $p^2 + 9p - 36$ **q** $x^2 + 9x - 52$ **r** $q^2 - 9q - 70$

PATH

EXAMPLE 24

5 Factorise each expression, given it is a perfect square. R

a	$m^2 + 4m + 4$	**b**	$p^2 + 20p + 100$	**c**	$a^2 - 10a + 25$
d	$y^2 - 14y + 49$	**e**	$w^2 + 24w + 144$	**f**	$a^2 - 3a + 2\frac{1}{4}$

6 Factorise each quadratic expression. Look for the highest common factor first. R

a	$3m^2 + 9m + 6$	**b**	$2y^2 + 2y - 4$	**c**	$5t^2 - 10t - 400$
d	$5e^4 + 25e^3 - 120e^2$	**e**	$x^3 - x^2 - 110x$	**f**	$4b^2 - 4b - 168$
g	$4w^2 + 4w - 48$	**h**	$3a^3 - 9a^2 - 12a$	**i**	$2e^2 + 18e + 40$
j	$24 - 5t - t^2$	**k**	$42 + u - u^2$	**l**	$28 + 3x - x^2$
m	$12 - b - b^2$	**n**	$7k - 12 - k^2$	**o**	$12x - 35 - x^2$

7 Factorise each quadratic expression. R

a	$h^2 - 3h - 18$	**b**	$18 - 7t - t^2$	**c**	$w^2 + 8w + 7$
d	$k^2 - 12k - 45$	**e**	$v^2 + 8v - 20$	**f**	$3c^2 + 30c + 75$
g	$q^2 - 6q + 5$	**h**	$a^2 - 4a + 3$	**i**	$6x^2 + 36x - 96$
j	$x^2 - 16x + 64$	**k**	$8u^2 - 24u - 32$	**l**	$b^2 + 11b + 30$
m	$y^2 - 12y + 36$	**n**	$5r^2 - 5r - 10$	**o**	$4l^2 - 8l - 32$
p	$g^2 - 24g + 80$	**q**	$108 - 6d - 2d^2$	**r**	$26 + 11n - n^2$

Investigation

Factorising quadratic trinomials by grouping in pairs

1 We can factorise the quadratic trinomial $6y^2 + 19y + 15$ by rewriting it as $6y^2 + 10y + 9y + 15$, splitting up the middle term $19y$.

- **a** Factorise $6y^2 + 10y + 9y + 15$ by grouping in pairs.
- **b** Factorise $6y^2 + 9y + 10y + 15$ by grouping in pairs. Is your answer the same as that for part **a**?
- **c** In $6y^2 + 19y + 15$, 6 is the coefficient of y^2 while 15 is the constant term. Find the product of 6 and 15 and the product of 10 and 9. What do you notice?

2 **a** Factorise $2c^2 + 13c + 21$ by rewriting it as $2c^2 + 6c + 7c + 21$.

- **b** Factorise $2c^2 + 13c + 21$ by rewriting it as $2c^2 + 7c + 6c + 21$.
- **c** Find the product of 2 and 21 and the product of 6 and 7. What do you notice?

1.13 Factorising quadratic expressions $ax^2 + bx + c$

PATH

Puzzle
Trinominoes

We will now factorise quadratic trinomials such as $6x^2 + 19x + 15$, where x^2 has a coefficient that is not a common factor. We do this by splitting the middle term into 2 terms and then factorise by grouping in pairs.

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Example 25

Factorise $3x^2 + 8x + 4$.

Solution

We need to split up the middle term $8x$.
Find 2 numbers that have a product of 12 and a sum of 8.

$3x^2 + 8x + 4 = 3x^2 + 8x + 4$

sum of 8

product of $3 \times 4 = 12$

The 2 numbers are $+6$ and $+2$, so we will split $8x$ into $6x$ and $2x$.

$\therefore 3x^2 + 8x + 4 = 3x^2 + 6x + 2x + 4$

$= 3x(x + 2) + 2(x + 2)$ Factorising by grouping in pairs

$= (x + 2)(3x + 2)$ Factorising again

Factorising quadratic trinomials of the form $ax^2 + bx + c$

- Find 2 numbers that have a sum of b and a product of ac
- Use these 2 numbers to split the middle term bx into 2 terms
- Factorise by grouping in pairs

Example 26

Factorise each quadratic expression.

a $5k^2 - 12k + 4$ **b** $9m^2 - 9m - 4$ **c** $6t^2 + t - 12$

Solution

a $5k^2 - 12k + 4$

$5 \times 4 = 20$

Find 2 numbers that have a product of 20 and a sum of -12.

$-5 \times (-4) = 20$; $-5 + (-4) = -9$ ✗

$-10 \times (-2) = 20$; $-10 + (-2) = -12$ ✓

Since the sum is negative and the product is positive, both of the numbers must be negative.

They are -10 and -2.

Split $-12k$ into $-10k$ and $-2k$.

$5k^2 - 12k + 4 = 5k^2 - 10k - 2k + 4$

$= 5k(k - 2) - 2(k - 2)$ Factorising by grouping in pairs

$= (k - 2)(5k - 2)$

b $9m^2 - 9m - 4$

$9 \times (-4) = -36$

Find 2 numbers with a product of -36 and a sum of -9.

Since the product is negative, one of the numbers must be negative.

$12 \times (-3) = -36$; $12 + (-3) = 9$ ✗

$-12 \times 3 = -36$, $-12 + 3 = -9$ ✓

PATH

Video
Factorising quadratic expressions (Advanced)

1.13

Video
Factorising quadratic expressions 2

PATH

They are –12 and 3.

$9m^2 - 9m - 4 = 9m^2 - 12m + 3m - 4$
$= 3m(3m - 4) + 1(3m - 4)$
$= (3m - 4)(3m + 1)$

c $6t^2 + t - 12$

$6 \times (-12) = -72$.

Find 2 numbers with a product of –72 and a sum of 1.

$8 \times (-9) = -72$; $8 + (-9) = -1$ ✗

$-8 \times 9 = -72$; $-8 + 9 = 1$ ✓

They are –8 and +9.

$6t^2 + t - 12 = 6t^2 - 8t + 9t - 12$
$= 2t(3t - 4) + 3(3t - 4)$
$= (3t - 4)(2t + 3)$

Example 27

Factorise each quadratic trinomial.

a $18a^2 - 18a - 8$ **b** $10 - 7x - 12x^2$

Solution

a $18a^2 - 18a - 8 = 2(9a^2 - 9a - 4)$ Taking out the HCF of 2 first
$= 2(9a^2 - 12a + 3a - 4)$ Product = –36, sum = –9
$= 2[3a(3a - 4) + 1(3a - 4)]$
$= 2(3a - 4)(3a + 1)$

b $10 - 7x - 12x^2 = -12x^2 - 7x + 10$ Rearranging the terms to make the x^2 term first
$= -(12x^2 + 7x - 10)$ Taking out a common factor of –1
$= -(12x^2 + 15x - 8x - 10)$ Product = –120, sum = –7
$= -[3x(4x + 5) - 2(4x + 5)]$
$= -(4x + 5)(3x - 2)$

EXERCISE 1.13 ANSWERS ON P. 611

Factorising quadratic expressions $ax^2 + bx + c$ UFR

EXAMPLE 25

1 Factorise each quadratic expression. R

a $2x^2 + 11x + 5$ **b** $4m^2 + 13m + 3$ **c** $5y^2 + 17y + 6$
d $6u^2 + 19u + 10$ **e** $2w^2 + 31w + 15$ **f** $4e^2 + 15e + 9$
g $8f^2 + 14f + 3$ **h** $3d^2 + 5d + 2$ **i** $2b^2 + 9b + 7$
j $5y^2 + 16y + 11$ **k** $8g^2 + 26g + 15$ **l** $6a^2 + 23a + 21$

EXAMPLE 26

2 Factorise each quadratic expression. R

a $2y^2 - 11y + 12$ **b** $10k^2 - 19k + 6$ **c** $6e^2 - 13e + 6$
d $4b^2 - 13b + 3$ **e** $6w^2 - 23w + 15$ **f** $12f^2 - 25f + 12$
g $15m^2 - 26m + 8$ **h** $9x^2 - 12x + 4$ **i** $2a^2 - 23a + 45$
j $12y^2 - 35y + 8$ **k** $8d^2 - 34d + 21$ **l** $4h^2 - 36h + 81$

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PATH

3 Factorise each quadratic expression. R

a $5y^2 - 6y - 11$	**b** $4d^2 - d - 5$	**c** $2m^2 - 3m - 9$
d $3t^2 - t - 30$	**e** $6h^2 - h - 7$	**f** $2y^2 - 5y - 12$
g $8a^2 - 2a - 3$	**h** $15u^2 - 7u - 4$	**i** $9c^2 - 12c - 5$
j $6c^2 - 7c - 24$	**k** $20n^2 - 27n - 14$	**l** $12x^2 - 7x - 10$

4 Factorise each quadratic expression. R

a $5m^2 + 2m - 7$	**b** $6g^2 + g - 12$	**c** $3p^2 + 4p - 4$
d $7w^2 + 6w - 1$	**e** $5y^2 + 14y - 3$	**f** $3n^2 + 10n - 8$
g $4b^2 + 9b - 9$	**h** $8m^2 + 10m - 3$	**i** $3x^2 + 2x - 16$
j $8u^2 + 26u - 15$	**k** $24e^2 + 31e - 15$	**l** $20h^2 + 11h - 42$

5 Factorise each expression given it is a perfect square. R

a $81w^2 - 180w + 100$	**b** $4y^2 + 8y + 4$	**c** $25h^2 - 40h + 16$

6 Factorise each quadratic expression by first taking out a common factor. R

a $6y^2 + 10y - 4$	**b** $6g^2 + 15g - 36$	**c** $24e^2 - 28e - 12$
d $8a^2 - 10a - 12$	**e** $12u^2 + 20u - 8$	**f** $-25q^2 - 5q + 6$
g $-12m^2 + 14m - 4$	**h** $20 - h - 12h^2$	**i** $18 + 48c + 24c^2$
j $15 + 9w - 6w^2$	**k** $12d^2 + 2d - 30$	**l** $22x - 12 - 6x^2$

7 Factorise each quadratic expression. R

a $2a^2 + 5a + 3$	**b** $12m^2 - 32m + 5$	**c** $4x^2 + 11x - 3$
d $7w^2 - 8w + 1$	**e** $4h^2 - 7h - 15$	**f** $8x^2 - 2x - 3$
g $5r^2 + 26r + 5$	**h** $2d^2 - 15d + 7$	**i** $6n^2 - 7n - 3$
j $8 - 6m - 9m^2$	**k** $3 - 2c - 5c^2$	**l** $15g^2 + 19g + 6$
m $15 + 14q - 8q^2$	**n** $3x^2 - 13x + 14$	**o** $16 - 8d - 3d^2$
p $42 - 10y - 12y^2$	**q** $28d^2 - 44d - 24$	**r** $100k^2 + 80k + 16$

1.14

EXAMPLE 27

Mixed factorisations

1.14

Factorisation strategies

- Look for any common factors and factorise first
- If there are 2 terms, try factorising using the difference of 2 squares
- If there are 3 terms, try factorising as a quadratic trinomial
- If there are 4 terms, try factorising by grouping in pairs

PATH

Worksheet
Mixed factorisations

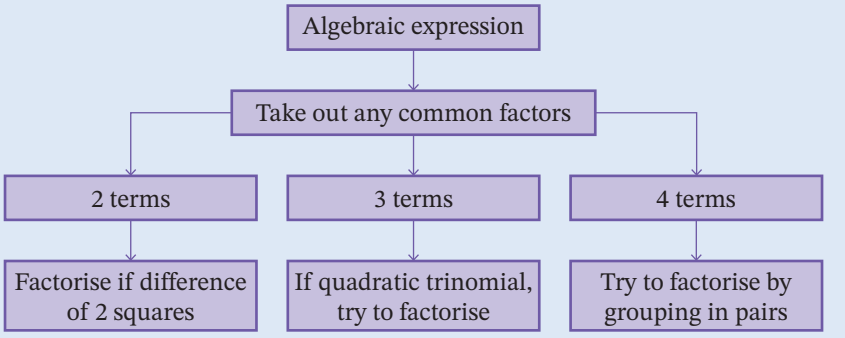
Puzzle
Factorising puzzle (Advanced)

Foundation | Standard | Complex

PATH

Puzzles
Factorominoes

Products and factors squaresaw



Video
Mixed factorisations

Example 28

Factorise each quadratic expression.

a $3a^2 - 27$ **b** $5a^2 + 100$ **c** $20b^2 - 52b + 24$ **d** $d^3 - d^2 - d + 1$

Solution

a $3a^2 - 27 = 3(a^2 - 9)$
$= 3(a + 3)(a - 3)$
Taking out the HCF of 3 first
Difference of 2 squares

b $5a^2 + 100 = 5(a^2 + 20)$
2 terms but not a difference of 2 squares

c $20b^2 - 52b + 24 = 4(5b^2 - 13b + 6)$
$= 4(5b^2 - 10b - 3b + 6)$
$= 4[5b(b - 2) - 3(b - 2)]$
$= 4(b - 2)(5b - 3)$

d $d^3 - d^2 - d + 1 = d^2(d - 1) - 1(d - 1)$
$= (d - 1)(d^2 - 1)$
$= (d - 1)(d + 1)(d - 1)$
$= (d - 1)^2(d + 1)$

EXERCISE 1.14 ANSWERS ON P. 611

Mixed factorisations U F R

EXAMPLE 28

1 Factorise each expression. R

a $m^2 - 16m + 64$ **b** $3d^2 - 3d$ **c** $3w^2 - 4w - 15$
d $3k - 15 - 5h + hk$ **e** $25y^2 - 64$ **f** $100f^2 - 64$
g $q^2 + 3q - 3pq$ **h** $3 + 2g - g^2$ **i** $24n^2 + 44n - 40$
j $25r^2 - 1$ **k** $b^3 + b^2 + b + 1$ **l** $4x^2 - 20x + 25$
m $4 - d - 5d^2$ **n** $a^3 - a^2 - a + 1$ **o** $8 - 2v^2$
p $mn^2 + mnp + 3mn + 3mp$ **q** $2w^2 - 24w + 72$ **r** $36h^2 + 12h + 1$

Foundation Standard Complex

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2 Factorise each expression. R

a	$15r^2 - 31rt - 24t^2$	**b**	$4d^2 + 4d + 1$	**c**	$9g^2 - 36k^2$
d	$e^3 - 3e^2 - 10e$	**e**	$5(p+q)^2 - 125(p-q)^2$	**f**	$28x^2 - 7$
g	$a^2 - b^2 + 4a - 4b$	**h**	$c^3 - 2c^2 - 4c + 8$	**i**	$6a^2 + 13a - 5$
j	$y^2 - 3y + 5y - 35$	**k**	$18p^2 + 24p + 8$	**l**	$1 - 2a - 24a^2$
m	$9x^2 - 27x + 18x - 54$	**n**	$2a^2b - 6ab - 3a + 9$	**o**	$2a^2 + 12a + 18$
p	$25u^2 - 10u + 1$	**q**	$4k^2 - 5k - 21$	**r**	$48 - 3w^2$
s	$3 - 27c^2$	**t**	$k^3 + 4k^2 - 16k - 64$	**u**	$5y^3 - 10y^2 + 15y$
v	$m^3n - 4mn$	**w**	$8 - 2a^2$	**x**	$32c^2 - 40c - 12$

3 Factorise each expression. R

a	$12e^2 + 13e - 35$	**b**	$10y^2 + 19ky - 15k^2$	**c**	$16h^4 - 81$
d	$w^2 - (6-w)^2$	**e**	$d^3 - 16d + 4 - d$	**f**	$u^2 - 8u + 16 - k^2$

PATH

1.15

Factorising algebraic fractions 1.15

Example 29

Simplify each expression.

a $\dfrac{10a+25b}{5}$ **b** $\dfrac{9y^2-16}{6y+8}$ **c** $\dfrac{x^2+x}{-4x-4}$ **d** $\dfrac{t^2-3t+2}{3t^2-5t-2}$

Solution

a $\dfrac{10a+25b}{5} = \dfrac{\cancel{5}^{\,1}(2a+5b)}{\cancel{5}^{\,1}}$ Factorise, then cancel.

$= 2a + 5b$

b $\dfrac{9y^2-16}{6y+8} = \dfrac{(3y+4)(3y-4)}{2(3y+4)}$

$= \dfrac{\cancel{(3y+4)}(3y-4)}{2\cancel{(3y+4)}}$

$= \dfrac{3y-4}{2}$

c $\dfrac{x^2+x}{-4x-4} = \dfrac{x(x+1)}{-4(x+1)}$

$= \dfrac{x\cancel{(x+1)}}{-4\cancel{(x+1)}}$

$= -\dfrac{x}{4}$

d $\dfrac{t^2-3t+2}{3t^2-5t-2} = \dfrac{(t-2)(t-1)}{(t-2)(3t+1)}$

$= \dfrac{\cancel{(t-2)}(t-1)}{\cancel{(t-2)}(3t+1)}$

$= \dfrac{t-1}{3t+1}$

PATH

Worksheets
Simplifying algebraic fractions
Algebra 2

Foundation Standard Complex

PATH

Example 30

Simplify each expression.

a $\dfrac{4}{x^2+x}+\dfrac{3}{x^2-1}$

b $\dfrac{3}{m^2+2m-3}-\dfrac{2}{m^2+2m-15}$

c $\dfrac{3m-6}{4}\times\dfrac{8m}{m^2-2m}$

d $\dfrac{d^2+3d+2}{d^2-9}\div\dfrac{d^2+d}{3d+9}$

Solution

a $\dfrac{4}{x^2+x}+\dfrac{3}{x^2-1}=\dfrac{4}{x(x+1)}+\dfrac{3}{(x+1)(x-1)}$

$=\dfrac{4(x-1)}{x(x+1)(x-1)}+\dfrac{3x}{x(x+1)(x-1)}$

$=\dfrac{4x-4+3x}{x(x+1)(x-1)}$

$=\dfrac{7x-4}{x(x+1)(x-1)}$

b $\dfrac{3}{m^2+2m-3}-\dfrac{2}{m^2-2m-15}$

$=\dfrac{3}{(m+3)(m-1)}-\dfrac{2}{(m+3)(m-5)}$

$=\dfrac{3(m-5)}{(m+3)(m-1)(m-5)}-\dfrac{2(m-1)}{(m+3)(m-1)(m-5)}$

$=\dfrac{3(m-5)-2(m-1)}{(m+3)(m-1)(m-5)}$

$=\dfrac{3m-15-2m+2}{(m+3)(m-1)(m-5)}$

$=\dfrac{m-13}{(m+3)(m-1)(m-5)}$

c $\dfrac{3m-6}{4}\times\dfrac{8m}{m^2-2m}=\dfrac{3(m-2)}{4}\times\dfrac{8m}{m(m-2)}$

$=\dfrac{3\cancel{(m-2)}}{\cancel{4}_1}\times\dfrac{\overset{2}{\cancel{8}}\,\cancel{m}}{\cancel{m(m-2)}}$

$=6$

d $\dfrac{d^2+3d+2}{d^2-9}\div\dfrac{d^2+d}{3d+9}=\dfrac{d^2+3d+2}{d^2-9}\times\dfrac{3d+9}{d^2+d}$

$=\dfrac{(d+2)(d+1)}{(d+3)(d-3)}\times\dfrac{3(d+3)}{d(d+1)}$

$=\dfrac{(d+2)\cancel{(d+1)}}{\cancel{(d+3)}(d-3)}\times\dfrac{3\cancel{(d+3)}}{d\cancel{(d+1)}}$

$=\dfrac{3(d+2)}{d(d-3)}$

EXERCISE 1.15 ANSWERS ON P. 612

Factorising algebraic fractions U F R

EXAMPLE 29

1 Simplify each expression. R

a $\dfrac{3x+3y}{3}$

b $\dfrac{5}{10t-10r}$

c $\dfrac{ab-ac}{a^2}$

d $\dfrac{y-1}{1-y}$

e $\dfrac{w^2-16}{w+4}$

f $\dfrac{5d-5w}{d^2-w^2}$

g $\dfrac{(k+5)^2}{k^2-25}$

h $\dfrac{6c^2-6}{2c+2}$

i $\dfrac{am-an+m-n}{m^2-n^2}$



Foundation Standard Complex

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j $\frac{y^2+9y+20}{2y+10}$ **k** $\frac{k^2-3k-4}{k^2-16}$ **l** $\frac{16a^2-25c^2}{4a^2-9ac+5c^2}$

m $\frac{x^2+4x+4}{x^2-x-6}$ **n** $\frac{1-c-2c^2}{3c^2+2c-1}$ **o** $\frac{ap+4p-2a-8}{2p^2-8}$

p $\frac{c^2-9c+20}{cu-4+c-4u}$ **q** $\frac{h^2+3h+2}{2h^2+4h}$ **r** $\frac{8e^2-14e-15}{4e^3-25e}$

2 Simplify each expression. R

a $\frac{5}{m(m+1)}+\frac{2}{(m+1)(m+2)}$ **b** $\frac{6}{(w+5)(w+3)}-\frac{4}{w(w+3)}$

c $\frac{3}{(b+2)(b-1)}+\frac{1}{(b-1)(b-3)}$ **d** $\frac{2}{k^2+k}-\frac{3}{k^2-1}$

e $\frac{5}{4h+4}+\frac{3}{h^2+h}$ **f** $\frac{3}{d^2+3d+2}-\frac{4}{d+2}$

g $\frac{3}{r^2-36}-\frac{5}{4r+24}$ **h** $\frac{3}{d^2+2d}+\frac{d}{d^2-4}$

i $\frac{5}{k^2-3k-4}-\frac{k}{k^2-1}$ **j** $\frac{2}{q^2-1}+\frac{3}{q+1}$

k $\frac{4}{a^2-8a+15}+\frac{3}{a^2+4a-21}$ **l** $\frac{2}{y^2+5y+6}-\frac{7}{y^2-y-12}$

m $\frac{5}{e^2+3e+2}-\frac{8}{e^2+9e+14}$ **n** $\frac{3}{6h^2-h-2}+\frac{4}{12h^2-23h+10}$

3 Simplify each expression. R

a $\frac{3m+9}{2}\times\frac{4m}{m+3}$ **b** $\frac{5d-10}{3d-9}\times\frac{5d-15}{8d-16}$ **c** $\frac{4}{e+2}\times\frac{e^2+2e}{8e}$

d $\frac{3k+6}{5}\times\frac{10k}{k+2}$ **e** $\frac{5h}{3h+9}\times\frac{6h+18}{h^2+h}$ **f** $\frac{4}{a^2-b^2}\times\frac{3a+3b}{8}$

g $\frac{r+t}{t^2-r^2}\times\frac{r^2-rt}{5r+5t}$ **h** $\frac{20m+16}{7m-7}\times\frac{7m}{5m+4}$ **i** $\frac{p^2+2p+1}{p^2-1}\times\frac{4p-4}{p^2+p}$

j $\frac{y+2}{5y}\div\frac{7y+14}{15y}$ **k** $\frac{5}{x^2-4}\div\frac{15}{2x+4}$ **l** $\frac{4n+8}{n+5}\div\frac{6n+12}{5n+25}$

m $\frac{d^2+d}{d+3}\div\frac{6d}{d^2-9}$ **n** $\frac{1}{f^2-6f+9}\div\frac{4}{f^2-9}$ **o** $\frac{3f+6}{f^2+f-6}\div\frac{f^2-2f-8}{f^2-f-12}$

PATH

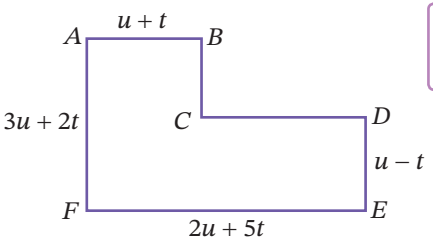
1.15

EXAMPLE 30

Foundation Standard Complex

Power plus ANSWERS ON P. 612

1 For this composite shape, write an expression for:



- **a** the length of CD
- **b** the length of BC
- **c** its perimeter
- **d** its area

2 Find the average of $6r$, $2r + 8$, $r - 5$, $2r$, $r + 7$ and $3r + 8$.

3 A rectangular garden has its longer sides each 5 m longer than its shorter sides. If its longer sides each have length y m, then write a simplified expression for:

- **a** the area of the garden
- **b** the perimeter of the garden

4 How many hours does it take a car to travel a distance of $20t^2$ kilometres at an average speed of $4t$ km/h?

5 Expand and simplify each expression.

- **a** $(a + b)(a + b + c)$
- **b** $(x - 2)(x + y - 3)$
- **c** $(p + 3)(p - 4)(p - 5)$
- **d** $(x - 1)(y - 1)(w - 1)$
- **e** $(ad - 3r)^2$
- **f** $(x^3 + 2)^2$

6 Factorise each expression.

- **a** $n^2 + 4mn + 4m^2$
- **b** $x^2 - 2xy + y^2$
- **c** $25x^2 - 40xy + 16y^2$
- **d** $5a^2 - 30ab + 45b^2$
- **e** $c^4 + 2c^2 + 1$
- **f** $8 - 7k^2 - k^4$
- **g** $x^4 - 1$
- **h** $(a + b)^2 - c^2$
- **i** $(p + q)^2 - (p - q)^2$

7 If $x^2 + bx + c = (x + p)(x + q)$, what are the signs of p and q if:

- **a** b and c are both positive?
- **b** b and c are both negative?
- **c** b is positive and c is negative?
- **d** b is negative and c is positive?

8 Simplify each expression.

- **a** $\frac{x-1}{4} + \frac{4x+2}{10}$
- **b** $\frac{2x+5}{6} + \frac{x-4}{3}$
- **c** $\frac{x}{3} - \frac{x+1}{5}$

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CHAPTER 1 REVIEW

Language of maths

area diagram	binomial	binomial product	coefficient
common factor	constant term	difference of 2 squares	distributive law
expand	expression	factor	factorise
formula	grouping in pairs	highest common factor (HCF)	like terms
perfect square	pronumeral	product	quadratic trinomial
simplify	sum	variable	

Quiz
Language of maths 1

Puzzle
Algebra crossword

1 Why is it possible to add or subtract terms like $6n$ and $4n$ but not $7x$ and $7y$?

2 In the expression $2x^2 - 3x + 6$, what is:

a the constant term? **b** the coefficient of x^2?

3 In what type of algebraic expression is the highest power of the variable equal to 2?

4 Explain what **product** means.

5 In algebra, what is the opposite of **expand?**

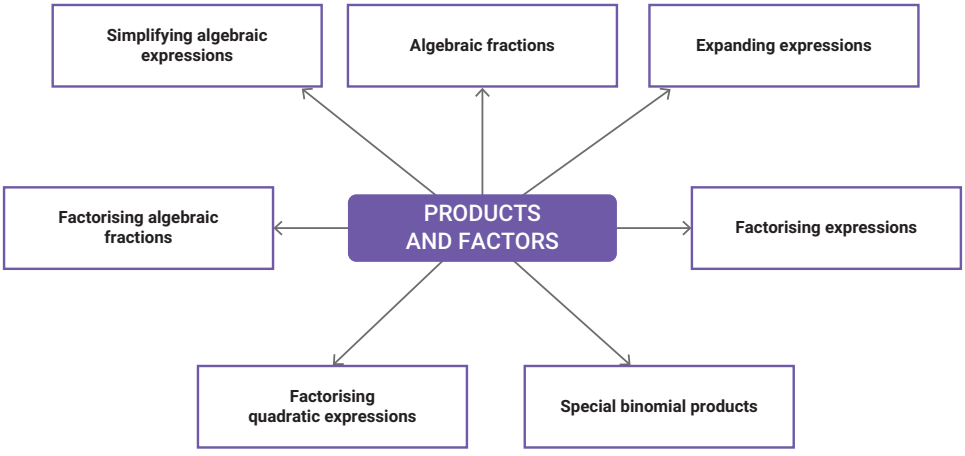
6 Complete: A binomial is an algebraic expression with ____ _____.

Topic summary

- Using your own words, write down the new things you have learnt.
- What parts of this topic did you like?
- What parts of this topic did you find difficult or you did not understand?
- Make a list of the skills you have learnt in this topic, such as substituting and expanding.

Worksheet
Mind map: Products and factors

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Quiz
Test yourself 1

TEST YOURSELF 1

ANSWERS ON P. 612

1.01

1 Simplify each expression.

a $9y + 7y$ **b** $4ab - 7ab$ **c** $3x^2 + 5x^2$

d $7p + 2q - 5p$ **e** $4w - 5 + 2w$ **f** $8x^2 - 13x - 5x^2$

g $2m + 3n + 5m + 7n$ **h** $5u + 7 - 2u - 3$ **i** $12d^2 + 5d - 7d + 4$

1.02

2 Simplify each expression.

a $9 \times 2n$ **b** $g \times 3h$ **c** $a \times d$

d $(-3) \times 2r$ **e** $12y^2 \div 3y$ **f** $2xy \div 8y$

g $20pq \div (-5q)$ **h** $15g^2 \div 25g$ **i** $36md^2 \div 27md$

1.03

3 Simplify each expression.

a $\frac{h}{6} + \frac{4h}{6}$ **b** $\frac{11k}{8} - \frac{7k}{8}$ **c** $\frac{7}{m} - \frac{5}{m}$

d $\frac{w}{4} - \frac{2w}{5}$ **e** $\frac{4k}{3} - \frac{k}{2}$ **f** $\frac{5}{2d} - \frac{4}{d}$

g $\frac{d}{5k} - \frac{3}{k}$ **h** $\frac{8}{m+1} - \frac{3}{m+1}$ **i** $\frac{2y}{5} + \frac{3d}{2}$

1.04

4 Simplify each expression.

a $\frac{5}{m} \times \frac{4}{i}$ **b** $\frac{h}{k} \times \frac{3}{k}$ **c** $\frac{14}{3y} \times \frac{9y}{2d}$

d $\frac{r}{4} \div \frac{r}{5}$ **e** $\frac{5}{v} \div \frac{2}{v}$ **f** $\frac{a}{b} \div \frac{a}{b}$

g $\frac{m}{5} \div \frac{5m}{10d}$ **h** $\frac{4}{x} \div \frac{12}{xp}$ **i** $\frac{ad}{6} \times \frac{5a}{2} \div \frac{ad}{8}$

j $\frac{3c}{4d} \div \frac{9c}{16d} \div \frac{6}{8}$ **k** $\frac{15a}{4b} \div \frac{4}{3a^2b} \times \frac{a}{bc}$ **l** $\frac{5y}{8} \div \frac{y}{4} \times \frac{15a}{2y}$

1.05

5 Expand and simplify each expression.

a $5(r - 2)$ **b** $a(a - 10)$ **c** $3y(2x - 5y)$

d $3p(4p - 7q)$ **e** $-6n(n - 9)$ **f** $-6t(t + 7)$

g $b(b + 1) + 5(b + 1)$ **h** $3x(2x - 7) - 5(2x - 7)$ **i** $12 - 6(2x - 3)$

j $5p(2 + 4p - 3q) - (p - q)$ **k** $5(2 - 4p - 3q) - (p + q)$

1.06

6 Factorise each expression.

a $4m - 12$ **b** $xy + xp$ **c** $-3g - 30$

d $16w^2 + 24w$ **e** $-8q + 16$ **f** $15k - 10hk^2$

g $3(y - 6) + y(y - 6)$ **h** $3x(2x - 1) - 2(2x - 1)$ **i** $p(p + 6) - 2(p + 6)$

1.07

7 Expand each binomial product.

a $(a + 2)(a + 3)$ **b** $(y + 3)(y - 7)$ **c** $(t - 3)(t + 8)$

d $(h - 5)(h - 4)$ **e** $(4 - g)(3 + g)$ **f** $(3p + 1)(2p + 3)$

g $(5x + 1)(4x - 3)$ **h** $(4r - 5)(3r + 4)$ **i** $(5 - 3q)(2q - 1)$

Foundation Standard Complex

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8 Expand each perfect square. 1.08

a $(n+5)^2$ **b** $(u-3)^2$ **c** $(9+x)^2$ **d** $(6-k)^2$

e $(3y+1)^2$ **f** $(2x-5)^2$ **g** $(6-7b)^2$ **h** $(1-4h)^2$

9 Expand each expression. 1.09

a $(n+6)(n-6)$ **b** $(10+m)(10-m)$ **c** $(3-8w)(3+8w)$

10 Expand and simplify each expression. 1.10

a $(2x+5)(x-1)+(x+3)^2$ **b** $(y+4)^2-(y-1)(y+2)$

c $(3a+1)(3a-1)-(3a+1)^2$ **d** $(2p-3q)^2+(2p+3q)^2$

11 Factorise each expression. 1.11

a $5mn-3n+10m-6$ **b** $ab-a-b+1$ **c** $x^2y+xy^3-5x-5y^2$

d $10p^2+30+p^3+3p$ **e** w^2-9 **f** m^2-n^2

g $36-r^2$ **h** $16a^2-9$ **i** $81x^2-y^2$

j $49a^2-25b^2$ **k** $3q^2-27$ **l** $50m^2-32$

m $24-150d^2$ **n** $4d^2+3de-3e^2-4de$ **o** $15h^2-6hk+2k^2-5kh$

12 Factorise each expression. 1.12

a $m^2-11m-12$ **b** y^2-4y+3 **c** $h^2-8h-65$

d $p^2+6p-27$ **e** $3n^2+9n+6$ **f** $4y^2+4y-8$

g $18+3x-x^2$ **h** $60-5b-5b^2$ **i** $11k-4-6k^2$

13 Factorise each expression. 1.13

a $6x^2+x-2$ **b** $5p^2-7p-6$ **c** $8w^2-14w+3$

d $3e^2+12e+12$ **e** $6y^2+15y-36$ **f** $24a^2-28a-12$

g $36x-27x^2-12$ **h** $52k-30-16k^2$ **i** $12g^2-46g+14$

14 Factorise each expression. 1.14

a $64y^2-25$ **b** $x^2-16x+64$ **c** $16r-48$

d $3g-15-5h+hg$ **e** a^3-a^2-a+1 **f** $3b^3-3b$

g $4y^2-5y-21$ **h** $18q^2+24q+8$ **i** $1-2p-24p^2$

15 Simplify each expression. 1.15

a $\dfrac{9y^2-25}{12y-20}$ **b** $\dfrac{2x^2+3x-5}{6x^2+19x+10}$ **c** $\dfrac{3w^2-75}{3w^2-30w+75}$

16 Simplify each expression. 1.15

a $\dfrac{4}{x^2+x}-\dfrac{3}{x^2-1}$ **b** $\dfrac{4}{x^2-9}+\dfrac{2}{3x+9}$

c $\dfrac{k}{2k+6}+\dfrac{5}{k^2-9}$ **d** $\dfrac{1}{(x-3)(x+2)}+\dfrac{4}{(x+2)(x+3)}$

e $\dfrac{4d}{d^2-3d+2}-\dfrac{3}{d^2-2d}$ **f** $\dfrac{3}{m^2-2m-3}-\dfrac{2}{m^2+5m+4}$

g $\dfrac{3d-6}{d+3}\times\dfrac{3d+9}{5d-10}$ **h** $\dfrac{y^2-5y+6}{y^3-4y}\times\dfrac{2y^2-8}{y^2-2y}$

i $\dfrac{4}{d^2-4}\div\dfrac{8}{d+2}$ **j** $\dfrac{m}{m^2+m}\div\dfrac{4}{5m+5}$

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